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## Full Length Articles

## Foreign currency loans and credit risk: Evidence from U.S. banks☆

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## ABSTRACT

When firms borrow in foreign currency and are not perfectly hedged, exchange rate changes can affect their ability to repay the debt. U.S. loan-level data show that a 10 percent depreciation of the local currency quarter-to-quarter increases the probability that a firm becomes past due on its loans by 42 basis points for firms with foreign currency debt relative to those with local currency debt. This increase is economically significant, given a baseline probability of 20 basis points, and indicates that exchange rate risk of borrowers can translate into credit risk for banks. Firms are more likely to borrow in foreign currency if they belong to industries that generate more income abroad and if a UIP deviation makes foreign currency loans cheaper. The paper establishes additional facts on large U.S. banks' international corporate loan portfolios, offering a perspective complementary to that provided by syndicated loan data.

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## 1. Introduction

Borrowing in foreign currency is a prevalent phenomenon, especially in emerging market economies, and has increased (CGFS, 2020). While foreign currency loans can be cheaper than local currency loans, they can expose firms to exchange rate risk by creating a currency mismatch on their balance sheets. Financial markets offer instruments to hedge against this risk, but hedging is costly and firms often remain unhedged.<sup>1</sup> As a result, firms may be negatively affected when the local currency depreciates.

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<sup>1</sup> On the trade-offs involved in hedging, see, for example, Nance et al. (1993) and Géczy et al. (1997). In a recent paper with detailed data on FX hedging in Chile, Alfaro et al. (2021) find that natural hedges are limited and financial hedges are concentrated in large firms.

The literature has emphasized the potential for this so-called balance sheet channel to counter or even overturn the classic expenditure-switching effect through which devaluations positively affect the performance of domestic firms (Mundell, 1968).<sup>2</sup> As policy makers set policies that often directly or indirectly move the exchange rate, it is crucial to better understand the effect of exchange rate changes on firm performance and, in particular, the strength of the balance sheet channel. Exploiting unique loan-level data from U.S. regulatory filings covering a large set of firms in 79 countries, this paper provides new evidence on the effect of exchange rate movements in the presence of foreign currency debt. Specifically, it analyzes how exchange rate changes affect firms' ability to service their debt depending on the currency in which they borrow.

We start by presenting a parsimonious model of currency choice and default with four testable predictions: (i) A local currency depreciation raises the default probability of a firm borrowing in foreign currency relative to a firm borrowing in local currency. (ii) This effect is smaller if the firm borrowing in foreign currency has a natural hedge. Also, firms are more likely to borrow in foreign currency (iii) if a larger share of their sales is in foreign currency and (iv) if there is a UIP deviation that reduces borrowing costs in foreign currency.

To test these predictions, we employ regulatory filings of large U.S. banks that participate in the Federal Reserve's Comprehensive Capital Analysis and Review (CCAR). The so-called Y-14 reports contain detailed quarterly information on banks' corporate loans and leases to non-U.S. borrowers with a loan amount of at least \$1 million originated by domestic and foreign offices.<sup>3</sup> We observe the location and the rating of the borrower, the currency in which the loan is denominated, and whether and for how long a loan has been past due. In contrast to other cross-country studies that rely on syndicated loan data, 82 percent of the loans in the dataset in this paper are not syndicated, implying that our dataset captures loans that are substantially smaller.<sup>4</sup> The sample includes around 26,000 borrowers around the world and runs from 2014q4 to 2018q4, a period that covers both a substantial USD appreciation and a notable period of USD depreciation.

We start by testing the effect of exchange rate changes on the past due status of firms (testable prediction (i)). To isolate the balance sheet channel from the classic expenditure switching channel, we employ a difference-in-difference type approach, comparing outcomes for firms in the same country, industry, quarter, and with the same bank-internal risk rating that borrow in different currencies. The strategy therefore controls for exchange rate effects through other channels (for example, through changes in demand) that are common across firms within these narrowly defined buckets. In addition, we control for the size of loans and their maturities.

In line with the theory, a 10 percent depreciation of the local currency quarter-to-quarter increases the probability that a firm that borrowed in foreign currency becomes past due on its loans by 42 basis points relative to a firm that borrowed in local currency. This effect mainly stems from local currency depreciations and is stronger for firms in industries with a smaller share of foreign sales, in line with testable prediction (ii). Since the baseline probability that a loan becomes past due is around 0.2 percent, the effect is economically meaningful.

It is also considerable given that our estimate should constitute a lower bound for the average treatment effect of borrowing in foreign currency under random assignment. As the theoretical model highlights, a firm's currency choice is not random but depends on the currency composition of its income. And while we compare firms in the same industry, country, and quarter and with the same bank-internal rating, loan size and maturity structure, these firms could still differ in their foreign income. We cannot control for this selection because of a lack of data, but selection implies that the estimated effects are biased towards zero because firms tend to borrow in the currency that offers a natural hedge: A firm with a revealed preference for local currency debt would be more exposed to exchange rate risk if it borrowed in foreign currency than a firm that actually decides to borrow in foreign currency.<sup>5</sup>

While the previously stated results are based on quarterly changes in the exchange rate, we also investigate whether exchange rate changes since loan origination matter for the past due status of firms' loans. Indeed, a 10-percent depreciation of the local currency from the date of origination until quarter  $t - 1$  increases the probability that a firm becomes past due on its loans in quarter  $t$  by 10 basis points for a firm that borrows in foreign currency relative to a firm that borrows in local currency.

Some firms that become past due on their loans will repay the loan the next period. For these firms, an exchange rate movement may merely create a liquidity problem, but we estimate that around 16 percent of the loans that become past due end up in default. We conclude that firms that borrow in foreign currency remain notably exposed to exchange rate risk, and, that as a result, the effects of exchange rate movements on firm default through the balance sheet channel are economically significant.

We also study how exchange rate changes affect the probability that a firm is downgraded or assigned a higher probability of default by its lender. In line with the results on the past due status of borrowers, firms that borrow in foreign currency are more likely to be downgraded or to be assigned a higher probability of default by the banks when the local currency depreciates than firms that borrow in local currency. Changes in banks' risk ratings/probabilities of default are much more frequent than the event

<sup>2</sup> Through the balance sheet channel, devaluations can be contractionary (Bebczuk et al., 2010; Kohn et al., 2015), cause or worsen currency crises (Aghion et al., 2001a; Aghion et al., 2004b; Ranciere et al., 2010), create systemic risk (Dell'Ariccia et al., 2016; Yesin, 2013), and magnify monetary policy spillovers (Akinci and Queralto, 2018). The potential for the balance sheet channel to dominate is high because, as recent work by Gopinath et al. (2016) and Bruno et al. (2018) shows, dominant currency pricing weakens the trade channel

<sup>3</sup> We cannot distinguish between loans originated by foreign offices and cross-border loans originated by domestic offices of U.S. banks, unfortunately.

<sup>4</sup> For example, Delis et al. (2018) study whether the exchange rate risk banks bear when they lend to borrowers in the currency of the borrower is passed on to borrowers through higher loan spreads using syndicated loan data from DealScan.

<sup>5</sup> We derive this selection effect formally in Section 2. In addition, Table 9 provides results consistent with earlier findings in the literature that firms are more likely to have foreign currency debt when they have foreign income or foreign assets (for example, Brown et al., 2011; Bleakley and Cowan, 2008; Kedia and Mozumdar, 2003; Keloharju and Niskanen, 2001).

that a firm becomes past due on its loans (3 and 7 percent of the sample versus 0.2 percent). This additional evidence can therefore address concerns associated with the low incidence of past-due borrowers in the sample.

The unique regulatory data also allow us to shed light on the factors that determine firms' currency choices. Foreign currency loans are larger and have shorter maturities than local currency loans. In line with the notion that firms choose their loan currency to match the currency composition of their income stream, firms in industries with a higher share of exports in total sales are more likely to borrow in foreign currency. Similarly, firms are more likely to borrow in foreign currency the cheaper the foreign currency loan is relative to the local currency loan in line with testable prediction (iv).<sup>6</sup> On average, foreign currency loans have a 100 to 140 basis point lower interest rate. If we adjust interest rates for expected future changes in the exchange rate to obtain the UIP deviations, the difference shrinks to 47–60 basis points.

This paper adds to a rich literature evolving around foreign currency debt and its macroeconomic effects.<sup>7</sup> The most closely related papers use micro-level data to study the effects on firm-level outcome variables. A number of papers consider effects on firm investment and employment, often focussing on crisis episodes. Aguiar (2005) shows that firms in Mexico with more short-term foreign debt exposure had substantially lower investments after a large devaluation.<sup>8</sup> Along these lines, Kim et al. (2015) report that firms' economic performance declined and default rates increased more for firms with foreign currency debt during the 1997–1998 Korean crisis. In contrast, Bleakley and Cowan (2008) do not find such differential effects, using accounting data for five Latin American countries.<sup>9</sup> Analyzing Mexican firms' responses to a large devaluation, Hardy (2018) finds that small firms invest less while large firms invest more. Kalemli-Ozcan et al. (2016) document adverse effects of devaluations on firm investment in the presence of foreign-denominated debt but only if there is a contemporaneous banking crisis.<sup>10</sup> Similar to this paper, Bruno and Shin (2020) and Alfaro et al. (2019) study the role of exchange rate fluctuations for corporate distress outside of crises, focussing on emerging markets. Bruno and Shin (2020) unveil that the stock prices of emerging market firms fall when the firms have outstanding dollar bonds and cash balances in local currency. Alfaro et al. (2019) discover that local currency depreciation increases the effect of leverage on emerging market firm fragility, in particular for firms in non-tradeable sectors. In contrast to previous work that uses firm balance sheet information, this paper employs loan-level information and focuses on new outcome variables: firms' past due status and the internal risk rating and probabilities of default they are assigned by their lenders. Besides highlighting that firms around the world remain exposed to material exchange rate risk, the paper documents how indirect exchange rate exposure of firms can translate into credit risk for banks.<sup>11</sup>

The link between exchange rate changes and credit risk has been emphasized by macroeconomic papers studying the causes of financial crises. For example, Kaminsky and Reinhart (1999) find that the collapse of the domestic currency exacerbates banking crises.<sup>12</sup> But there is relatively little work on the topic in the micro-banking literature. Kim et al. (2016) study whether banks charge higher interest rates when the borrower is exposed to exchange rate risk. Božović et al. (2009) provide a model where exchange rate risk spills over into default risk, resulting in reduced credit supply and growth. Bruno and Shin (2015a) focus on the implications of local currency depreciation for the credit risk of global banks, showing empirically that banks reduce their cross-border lending when the U.S. dollar appreciates.<sup>13</sup>

The paper also contributes to the work that investigates the determinants of firms' currency choices (Brown et al., 2011, Bleakley and Cowan, 2008, Kedia and Mozumdar, 2003, and Keloharju and Niskanen, 2001).<sup>14</sup> Galindo et al. (2003) provides a survey on the determinants of debt currency denomination. In comprehensive work that is both theoretical and empirical, Varela and Salomao (2016) build a dynamic model of currency choice where heterogeneous firms face a tradeoff between exchange rate exposure and growth since cheaper foreign currency lending allows firms to grow faster.<sup>15</sup>

The remainder of the paper is structured as follows. Section 2 proposes a theoretical model of currency choice and default. Section 3 introduces the data sources and presents key facts about foreign currency loans in comparison to local currency loans. Section 4 presents the regression results. Section 5 concludes.

## 2. Theory

To provide a theoretical background for the subsequent empirical analysis, this section presents a parsimonious model of firm borrowing with endogenous currency choice.<sup>16</sup> The model delivers four predictions regarding the effect of exchange rates on firm

<sup>6</sup> Varela and Salomao (2016) also find that UIP deviations increase the fraction of firms that borrow in foreign currency using Hungarian firm-level data. Similarly, Basso et al. (2011) find that interest rate differentials matter for foreign currency borrowing (and lending).

<sup>7</sup> Early studies focus on macro-level analysis. See Edward (1986), Céspedes (2005) and Bebczuk et al. (2010), for example.

<sup>8</sup> Similarly, Carranza et al. (2003), Echeverry et al. (2003), Pratap et al. (2003), Cowan et al. (2005), and Cowan et al. (2006) find that firms with higher foreign debt contract investment more after devaluations.

<sup>9</sup> Several other papers do not find evidence for the balance channel, for example, Benavente et al. (2003), Bonomo et al. (2003), and Alvarez and Hansen (2017).

<sup>10</sup> Using bond data to identify firms' foreign currency exposures, Caballero (2018) finds that local currency depreciations lower firms' capital expenditures.

<sup>11</sup> We do not know the extent to which banks might hedge the indirect exchange rate exposure in their loan books. Regardless, the indirect exchange rate exposure does not appear to be substantial for banks, even in the absence of hedging, because foreign currency lending represents only a small portion of the banks' lending activity and the total number of loans that become past due each quarter is small.

<sup>12</sup> A recent paper, Weiss (2020), shows how currency news shocks in the 19th century affected corporate bond returns in the United States.

<sup>13</sup> For papers on banks' choices of lending currencies, see Ize and Yeyati (2003), Luca and Petrova (2008), and Brown and De Haas (2012a).

<sup>14</sup> For more recent studies, see Kamil (2012), Brown et al. (2014), Basso et al. (2011), and Keller (2018).

<sup>15</sup> In line with results in Varela and Salomao (2016), Kalemli-Ozcan et al. (2019) find that firms in countries with greater shares of FX debt in total debt reduce their leverage more than firms in countries with low FX debt when the local currency depreciates.

<sup>16</sup> The model is simple and, in spirit, similar to Varela and Salomao (2016), for example. Bruno and Shin (2015b) also model balance sheet effects proposing a double-decker model of the global banking system.

default and firms' optimal choices. First, a depreciation of the local currency weakly increases (decreases) the default probability of a firm that borrows in foreign (local) currency. Second, foreign currency income provides a natural hedge against this risk. Third, firms borrow in the currency in which they generate most of their income. Finally, if there is a deviation from uncovered interest parity (UIP), then firms borrow in the currency with the lower expected interest rate.

## 2.1. Basic setup

### 2.1.1. Firm problem

A firm borrows one unit of capital from a bank. It invests the capital in a project that delivers safe return  $R > 1$ . A fraction  $\alpha$  of that return is generated in local currency and a fraction  $1 - \alpha$  in foreign currency. The firm operates under limited liability, that is, the shareholders are residual claimants after bank loans are repaid but are not liable for any losses. The firm can choose the currency of its loan. Let  $r > 1$  and  $r^* > 1$  be the interest rates paid by a firm for a local and foreign currency loan, respectively.<sup>17</sup> Without loss of generality, normalize the exchange rate to 1 when the loan is issued. If a firm borrows in local currency, this implies the following profits in local currency:

$$\Pi^{\text{LC}} = \max \{0, (\alpha R - r) + s(1 - \alpha)R\}, \quad (1)$$

where the max operator reflects the fact that the firm operates under limited liability and cannot have negative profits. Note that if a firm has a local currency loan and some foreign income ( $\alpha < 1$ ), profits in local currency increase in the exchange rate  $s$ . Rearranging the equation, we can derive a minimum exchange rate that is required for a firm to be able to repay its loan:<sup>18</sup>

$$s \geq \frac{r - \alpha R}{(1 - \alpha)R} = \underline{s}. \quad (2)$$

Alternatively, a firm can borrow in foreign currency. Then, the firm's profits, expressed in local currency units, are given by:

$$\Pi^{\text{FC}} = \max \{0, \alpha R + s((1 - \alpha)R - r^*)\}. \quad (3)$$

If foreign currency income is sufficiently small ( $1 - \alpha < \frac{r^*}{R}$ ), profits in foreign currency decrease in the exchange rate  $s$ . In this case, a firm defaults on its loan for any value of the exchange rate  $s$  that is above  $\bar{s}$ :

$$s > \frac{\alpha R}{r^* - (1 - \alpha)R} = \bar{s}. \quad (4)$$

### 2.1.2. Default risk and exchange rates

For our empirical analysis, we want to understand how exchange rate changes affect default and how effects differ across firms that borrow in different currencies. The basic relationship between exchange rates and defaults is captured in the following proposition:

#### Proposition 1. Loan default and exchange rates

When the local currency depreciates ( $s$  increases), the probability that a firm defaults on its loan

- (i) Decreases when the loan is in local currency and the firm has limited local currency income ( $\alpha < \frac{r}{R}$ ).
- (ii) Increases when the loan is in foreign currency and the firm has limited foreign currency income ( $1 - \alpha < \frac{r^*}{R}$ ).
- (iii) Remains unchanged if
  - the loan is in local currency and the firm has sufficient local currency income ( $\alpha \geq \frac{r}{R}$ ),
  - or the loan is in foreign currency and the firm has sufficient foreign currency income ( $1 - \alpha \geq \frac{r^*}{R}$ ).

**Proof.** Follows directly from Eqs. (1) and (3). □

The proposition states that the exchange rate affects credit risk if there is sufficient currency mismatch at the firm level, that is, if there are levels of the exchange rate at which a firm is unable to pay back its loans. This is the case if a firm borrows in local currency but does not have enough local income ( $\alpha < \frac{r}{R}$ ) or if a firm borrows in foreign currency without sufficient foreign income ( $1 - \alpha < \frac{r^*}{R}$ ).

<sup>17</sup> To simplify notation,  $r$  and  $r^*$  include the principal.

<sup>18</sup> Note that in the baseline model, the only reason for a firm to default is an unfavorable realization of the exchange rate. In appendix E, we show how the model can be generalized to include an additional shock to profits.

We are particularly interested in the relative default risk of two firms that borrow in local and foreign currency, respectively, and may generate different shares of their income in local currency (different  $\alpha$ ). Suppose firm A borrows in foreign currency with local currency income share  $\alpha^{FC}$ . Firm B borrows in local currency and has local currency income share  $\alpha^{LC}$ . We can then prove the following corollary.

**Corollary 1. Relative default**

Suppose there are two firms that borrow in local and foreign currency, respectively, generating fractions  $\alpha^{LC}$  and  $\alpha^{FC}$  of their total income at home. When the local currency depreciates:

- (i) The default probability of the firm that borrows in foreign currency relative to the firm that borrows in domestic currency increases:
  - (a) If the firm borrowing in foreign currency faces default risk ( $1 - \alpha^{FC} < \frac{r^*}{R}$ ).
  - (b) If the firm borrowing in local currency faces default risk ( $\alpha^{LC} < \frac{r}{R}$ ).
- (ii) The relative default probability remains unchanged if  $1 - \alpha^{FC} \geq \frac{r^*}{R}$  and  $\alpha^{LC} \geq \frac{r}{R}$ .

**Proof.** Follows directly from Proposition 1. □

There are two cases. In the first case, a local depreciation increases the relative default risk of the firm that borrows in foreign currency. This happens if at least one of the firms faces exchange-rate related default risk ( $1 - \alpha^{FC} < \frac{r^*}{R}$  or  $\alpha^{LC} < \frac{r}{R}$ ). In the second case, a local depreciation has no effect on default rates. Then, firms have sufficient income in the currency of their loan and do not face any exchange-rate related default risk.

Corollary 1 implies that independent of the shares of foreign income generated,  $\alpha^{FC}$  and  $\alpha^{LC}$ , the probability of default of a firm borrowing in foreign currency weakly increases relative to that of a firm borrowing in local currency when there is a local depreciation. If a firm borrowing in foreign currency has a larger share of income in foreign currency than a firm borrowing in local currency (that is, if they borrow in line with their natural hedge, which is the case as shown in Section 2.2), then this relative effect becomes smaller and is either positive or zero but never negative.

Next we show how foreign currency income can serve as a natural hedge for a firm that borrows in foreign currency. Recall that a firm borrowing in foreign currency defaults whenever the exchange rate is above  $\bar{s}$  while a firm borrowing in local currency defaults whenever the exchange rate falls below  $\underline{s}$ . Probabilities of default are therefore given by:

$$PD^{FC} = \int_{\bar{s}}^{\infty} f(s)ds, \quad PD^{LC} = \int_0^{\underline{s}} f(s)ds. \quad (5)$$

Now, taking the derivatives with respect to  $\alpha$  delivers:

$$\frac{\partial PD^{FC}}{\partial \alpha} = -f(\bar{s}) \frac{R(r^* - R)}{(r^* - (1 - \alpha)R)^2} \geq 0, \quad \frac{\partial PD^{LC}}{\partial \alpha} = f(\underline{s}) \frac{R(r - R)}{((1 - \alpha)R)^2} \leq 0. \quad (6)$$

Our findings on the natural hedge are summarized in the following proposition:

**Proposition 2. Natural Hedge**

A firm's probability of default

- (i) Rises in the domestic sales share  $\alpha^{FC}$  if the firm borrowed in foreign currency.
- (ii) Falls in the domestic sales share  $\alpha^{LC}$  if the firm borrowed in domestic currency.

**Proof.** Follows directly from Eq. (6). □

The proposition is quite intuitive. If more income is in the currency of the loan, this income provides a natural hedge and any swing in the exchange rate creates less default risk for the firm.

## 2.2. Optimal currency choice

We now turn to the optimal currency choice of the firm. A firm's expected profits when borrowing in the local currency are:

$$E[\Pi^{LC}] = \int_{\underline{s}}^{\infty} [(\alpha R + s(1 - \alpha)R) - r]f(s)ds. \quad (7)$$

Expected profits from borrowing in foreign currency are:

$$E[\Pi^{\text{FC}}] = \int_0^s [\alpha R + s((1-\alpha)R - r^*)]f(s)ds. \quad (8)$$

When is a foreign currency loan preferred over a local currency loan? To answer this question, we calculate the difference between expected profits from these two options:

$$\begin{aligned} \Delta\Pi &= E[\Pi^{\text{FC}}] - E[\Pi^{\text{LC}}] \\ &= \int_0^s [\alpha R + s((1-\alpha)R - r^*)]f(s)ds - \int_s^\infty [(\alpha R + s(1-\alpha)R - r)]f(s)ds. \end{aligned} \quad (9)$$

The choice crucially depends on the interest rates charged to the firms,  $r$  and  $r^*$ , which are endogenous objects that we derive next.

### 2.2.1. Bank problem

Assume a perfectly competitive banking sector. Banks can borrow at the risk-free rates  $i > 1$  and  $i^* > 1$  in local and foreign currency, respectively. Banking markets for foreign and local currency are segmented, and banks cannot take on currency risk themselves; that is, for lending in foreign currency, they have to finance the exact loan amount at the foreign risk-free rate. If a firm defaults, a bank can recover a fraction  $\gamma \in [0, 1]$  of the revenues. In addition, assume that the local risk-free rate differs from the foreign risk-free rate by a factor  $\varphi > 0$ , so that  $i = \varphi i^*$ . Perfect competition implies that the bank's expected profits will be equal to the risk-free rate:

$$i = \int_s^\infty r f(s)ds + \int_0^s \gamma [\alpha R + s(1-\alpha)R]f(s)ds, \quad (10)$$

where the first integral reflects full repayment by the firm and the second integral captures the residual claim of the bank when the firm defaults. For a foreign currency loan, the bank needs to recover the foreign risk-free rate. We can write the bank's expected profits from a foreign currency loan in local currency units as:

$$si^* = \int_0^s sr^* f(s)ds + \int_s^\infty \gamma [\alpha R + s(1-\alpha)R]f(s)ds. \quad (11)$$

Rewriting Eq. (10) and combining it with Eq. (11) and plugging it into Eq. (9) delivers:

$$\begin{aligned} \Delta\Pi &= (\varphi - 1) \int_0^s sr^* f(s)ds + (1-\gamma) \int_0^s [\alpha R + s((1-\alpha)R)]f(s)ds \\ &\quad + (\varphi\gamma - 1) \int_s^\infty [\alpha R + s((1-\alpha)R)]f(s)ds. \end{aligned} \quad (12)$$

For the key insights from the model, consider three special cases.

*Case 1:*  $\gamma = 1, \varphi = 1$ . When there is no UIP deviation ( $\varphi = 1$ ) and banks can fully recover revenues in case of default, firms are indifferent between FC and LC loans, that is  $\Delta\Pi = 0$ . In particular, the currency choice is independent of the local currency income share  $\alpha$ .

*Case 2:*  $\gamma < 1, \varphi = 1$ . Next, consider the case without a UIP deviation but with costly default. Plugging in the corresponding parameter values delivers:

$$\Delta\Pi = (1-\gamma) \left( \int_0^s [\alpha R + s(1-\alpha)R]f(s)ds - \int_s^\infty [(\alpha R + s(1-\alpha)R)]f(s)ds \right). \quad (13)$$

Taking the derivative with respect to  $\alpha$ , we obtain:<sup>19</sup>

$$\frac{\partial \Delta\Pi}{\partial \alpha} = (1-\gamma)R \left( \int_0^s (1-s)f(s)ds - \int_s^\infty [(1-s)f(s)]ds \right) \geq 0. \quad (14)$$

When firm liquidation is costly for banks ( $\gamma < 1$ ), a firm is less likely to choose a foreign currency loan when more of its income is in local currency (higher  $\alpha$ ). This is intuitive: When default is costly, there is an incentive to pick the currency that, all else equal,

<sup>19</sup> Note that there are also derivatives with respect to the boundaries, following Leibniz' rule, but they are equal to zero as profits at the boundaries are zero.



minimizes default risk. The fact that firms optimally choose the currency of the loan so that they are naturally hedged against exchange rate moves implies  $\alpha^{LC} \geq \alpha^{FC}$ : A firm that selects a local currency loan has a weakly higher local currency income share than a firm that selects a foreign currency loan.

Case 3:  $\gamma = 1$ . Consider the case where banks can fully recover revenues in default but there is a UIP deviation:

$$\Delta\Pi = (\varphi - 1) \left( \int_s^\infty [\alpha R + s(1 - \alpha)R]f(s)ds + \int_0^s sr^*f(s)ds \right). \quad (15)$$

If  $\varphi > 1$ , the foreign risk-free rate is lower than the local risk-free rate, and firms, all else equal, prefer foreign currency loans. We summarize our findings in the following proposition.

**Proposition 3. Currency choice.** *A firm*

- (i) *Is indifferent between foreign currency and local currency loans if there is no UIP deviation and no liquidation cost ( $\varphi = 1$ ,  $\gamma = 1$ ).*
- (ii) *Chooses the currency of the loan in line with its natural hedge (implying  $\alpha^{LC} \geq \alpha^{FC}$ ) if there is no UIP deviation but there is a liquidation cost ( $\varphi = 1$ ,  $\gamma < 1$ ).*
- (iii) *Chooses the currency with the lower risk-free rate if there is a UIP deviation but no liquidation cost ( $\varphi \neq 1$ ,  $\gamma = 1$ ).*

**Proof.** Follows from Eqs. (12), (14) and (15). □

The key predictions relevant for our empirical analysis are (ii) and (iii). The former implies that firms tend to select the currency of the loan so as to minimize the default risk and, hence, tend to match the loan currency to the currency in which the income is denominated.<sup>20</sup> The latter states that firms select the currency in which it is cheaper to borrow, accounting for expected exchange rate changes.

### 2.3. Additional aspects

We briefly discuss some additional aspects that are not explicitly modeled.

#### 2.3.1. Financial hedges

If firms are not fully hedged through their revenue streams, they can buy protection against local currency depreciation or appreciation and engage in foreign exchange swaps.<sup>21</sup> However, not all firms may hedge because it is costly. Firms may face both a variable cost associated with individual hedges and a fixed overhead cost of participating in the foreign exchange market. Hedging with financial instruments should reduce the effect of local depreciations on default similar to the effect of natural hedges studied in the model.<sup>22</sup>

#### 2.3.2. Pass-through of debt servicing costs

If firms have a currency mismatch and debt servicing costs rise because the local currency depreciates, firms might pass on the higher cost to their customers through higher prices. However, firms may not be able to increase prices, either because of the market structure and competitive pressures or because prices are sticky in the short run. As the local currency depreciates, the cost of debt for these firms, which often face monthly interest payments, rises but prices cannot be adjusted promptly to compensate the firms for the higher cost. A large literature documents short-term price stickiness and less than perfect pass-through of higher costs to consumers.<sup>23</sup> To the extent that firms have some ability to adjust prices, for example, because competitors have similar foreign-currency exposures, the effect of local currency depreciations on defaults should be mitigated.

### 2.4. Key takeaways from the model

To summarize, the model developed in this section has four testable predictions.

#### Testable prediction (1)

A local depreciation raises the default probability of a firm that borrows in foreign currency relative to a firm that borrows in local currency.

<sup>20</sup> This result has found empirical support: Brown et al. (2009) report that foreign currency income is the dominant reason for foreign currency borrowing in Eastern Europe. Similarly, Bleakley and Cowan (2008), Kedia and Mozumdar (2003), and Keloharju and Niskanen (2001) find that firms obtain foreign currency debt to hedge against foreign currency income.

<sup>21</sup> Géczy et al. (1997) show that firms with foreign exchange rate exposures are more likely to use currency derivatives.

<sup>22</sup> It would be straightforward to extend the model in this direction by allowing a firm to vary its domestic currency share  $\alpha$  at a cost. Buying a financial hedge is then isomorphic to shifting income from one currency stream to the other at the cost of reducing expected income.

<sup>23</sup> See, for example, Klenow and Kryvtsov (2008), Klenow and Malin (2010), Nakamura and Steinsson (2013), and Gopinath and Rigobon (2008).

*Testable prediction (2)*

A local depreciation raises (lowers) the default probability by less if there is a smaller currency mismatch, that is the firm borrowing in foreign (local) currency has more sales in foreign (local) currency.

*Testable prediction (3)*

Firms are more likely to borrow in foreign currency if a larger share of their sales is in foreign currency.

*Testable prediction (4)*

Firms are more likely to borrow in foreign currency if there is a UIP deviation that reduces borrowing costs in the foreign currency.

The focus of this paper is on testable prediction 1, as we are interested in understanding how exchange rate changes affect the probability of default of firms that borrow in foreign currency. That said, we provide evidence in support of all four predictions in this paper. Of note, testable prediction 1 holds even when firms endogeneously choose the currency in which they borrow. As discussed above, under selection into currency, the effect of local depreciations on firms' relative default risk is dampened but not overturned.

### 3. Data sources and facts

This section provides information on data sources as well as some key facts about U.S. bank lending to foreign borrowers, including the incidence of loans becoming past due.

#### 3.1. Data sources

##### 3.1.1. Corporate loan data source

The loan-level data used in this paper come from Y-14 reports that the largest U.S. banks that are stress-tested by the Federal Reserve have to file on a quarterly basis.<sup>24</sup> Banks report corporate loans and leases that are held for investment and held for sale with a committed exposure above \$1 million.<sup>25</sup> They report at the consolidated level, that is, we observe not only cross-border loans extended to foreign firms by the parent bank but also those extended by the banks' foreign subsidiaries and branches (although we cannot distinguish between them). Reporting of the loan-level information started in 2011q3. However, information on the currency denomination of the loan, crucial for our analysis, is only available from 2014q4 onwards. Therefore, the baseline sample covers a shorter time period, running from 2014q4 to 2018q4. The sample includes 29 internationally active banks, some of which enter the sample as they become part of the annual stress-testing exercise. The five most international banks extend, on average, loans to borrowers in 63 countries, while the five least international banks lend, on average, to firms in 3 countries.

To obtain a consistent dataset, we subject the data to several cleaning procedures and collapse the loan-level dataset to the borrower level.<sup>26</sup> A borrower is identified as a combination of customer identifier, location, and bank. When collapsing the data, we calculate utilized exposure-weighted averages for all variables.<sup>27</sup> Further details on data cleaning can be found in the data appendix. The least restrictive sample has 147,103 observations and covers 26,345 borrowers residing in 79 countries.<sup>28</sup> 35 percent of the observations are for borrowers in the top 5 countries, which are Mexico, Canada, the UK, Brazil, and India.

##### 3.1.2. Additional data sources

The borrower-level dataset is complemented with data from a variety of sources. Information on bilateral exchange rates are from the IMF's International Financial Statistics. Country-industry level variables (specifically, international sales over total sales) are constructed from Worldscope balance sheet data. Further details on variables and their sources can be found in the data appendix.

#### 3.2. Characteristics of international loans

The Y-14 data contain unique information on the international corporate loan portfolios of the largest U.S. banks. In the following, we document key facts about these loan portfolios. The statistics provide novel insights into U.S. banks' foreign currency lending, as existing cross-country research is either based on more aggregated data (for example, on the BIS International Banking

<sup>24</sup> For a list of the stress-tested banks, see: <https://www.federalreserve.gov/publications/comprehensive-capital-analysis-and-review-summary-instructions-2020.htm>.

<sup>25</sup> Banks report corporate loans on schedule H.1. The data are confidential but available to researchers within the Federal Reserve System.

<sup>26</sup> The decision to delay payments on individual loans is taken at the borrower-level, as exchange rate shocks should operate at that level. In the data, most borrowers become past due on all their loans simultaneously (about 75 percent) and, on average, borrowers become past due on 86 percent of their utilized exposure.

<sup>27</sup> Utilized exposure represents the money that was disbursed to the firm by the bank, in contrast to committed exposure, which is the amount the bank has committed to pay out.

<sup>28</sup> We drop countries whose currencies are pegged to the USD as well as the United States. In addition, we drop observations for Ghana and Venezuela which had extremely volatile exchange rates over the sample period.



**Table 1**

Loan characteristics, by currency.

|                              | (1)    |      | (2)         |      | (3)    |      | (4)  |      |
|------------------------------|--------|------|-------------|------|--------|------|------|------|
|                              | USD    |      | oth foreign |      | local  |      | mix  |      |
|                              | mean   | p50  | mean        | p50  | mean   | p50  | mean | p50  |
| Util. Exposure (USD million) | 31.0   | 8.73 | 35.4        | 9.80 | 17.9   | 4.29 | 51.2 | 26.6 |
| Maturity (years)             | 5.82   | 4.76 | 5.74        | 4.99 | 5.22   | 4.68 | 6.21 | 5.00 |
| Prob. of default (pct)       | 1.79   | 0.66 | 1.42        | 0.64 | 2.00   | 0.71 | 1.58 | 0.66 |
| Interest rate (pct)          | 2.97   | 2.42 | 2.56        | 1.84 | 4.05   | 3.29 | 3.29 | 2.65 |
| Loans newly past due (pct)   | 0.27   | 0    | 0.23        | 0    | 0.18   | 0    | 0.13 | 0    |
| Observations                 | 74,286 |      | 6024        |      | 58,639 |      | 8154 |      |

The table shows summary statistics of the baseline sample with 147,103 observations grouped by the currency denomination of the loans. Column 1 includes observations where the borrower has loans denominated exclusively in USD. Column 2 has observations where the borrower has loans in a foreign currency other than the USD. Column 3 is based on borrowers with local currency loans. Column 4 includes observations where the borrowers have loans in multiple currencies. The table displays the means and the medians of the following variables: utilized exposure, maturity, bank-internal probability of default, interest rate, and a dummy variable which is one when the borrower becomes past due on its loans in period  $t$ .

Statistics) or on syndicated loan data. Syndicated loans, as shown below, represent a relatively small fraction of total international C&I loans on U.S. banks' balance sheets.<sup>29</sup> All statistics discussed below are calculated based on the loans of U.S. banks to foreign borrowers in the Y-14 data set. Importantly, they do not capture these firms' borrowings from other sources, for example, from local banks that are not subsidiaries of U.S. banks.

### 3.2.1. Foreign currency loans are prevalent, tend to be larger, and have similar ratings to local currency loans

50 percent of firm-quarter observations are accounted for by borrowers that exclusively borrow in USD while 40 percent of the firms borrow in local currency. 4 percent have loans in a foreign currency other than the USD, and 6 percent borrow in more than one currency. Table 1 shows the average and median sizes, maturities, probabilities of default, and interest rates of loans in different currencies. The table documents that firms that borrow in foreign currency take out larger loans than firms that borrow only in the local currency. Interestingly, as Fig. D.1 highlights, the distribution of loans across rating buckets does not differ much by currency.

### 3.2.2. Local currency loans carry a higher interest rate

As Table 1 indicates, local currency loans pay higher interest than foreign currency loans. To further explore this difference, we look at the sample of newly originated loans and regress the interest rate of a loan on a dummy variable that is one when the loan is denominated in USD, controlling for a battery of fixed effects as well as the size of the loan and its maturity. Results are shown in Table 2. All regressions include country-time and ratings fixed effects.<sup>30</sup> Columns (1) through (4) show results for fixed-rate loans while columns (5) through (8) show results for variable-rate loans. As columns (1) and (5) show, the interest rate of a foreign-currency loan is on average between 100 and 140 basis points lower than the interest rate of a comparable local currency loan. This interest rate difference partly reflects expectations about future moves in the bilateral exchange rate. To control for this effect, we adjust the interest rate for the expected change in the USD exchange rate vis-a-vis the local currency.<sup>31</sup> Differences in these adjusted interest rates between dollar loans and foreign currency loans indicate deviations from uncovered interest rate parity (UIP).<sup>32</sup> Indeed, columns (3) and (7) show that even when accounting for exchange rate expectations, foreign currency loans are between 47 to 60 basis points cheaper than domestic currency loans.

### 3.2.3. Volatile currencies have large UIP deviations

One reason why UIP may not hold is the cost of hedging against exchange rate fluctuations. That is, a U.S. bank may charge a higher interest rate on a foreign currency loan because it either takes on exchange rate risk or needs to pay for a hedge. We would therefore expect the UIP deviation to be larger for countries with more volatile exchange rates. Results in line with this hypothesis are presented in columns (2), (4), (6), and (8) of Table 1. For loans to firms in countries with above median exchange rate volatilities, the average interest rate difference between a USD loan and a foreign currency loan doubles, both for the observed interest rates (columns (2) and (6)) and for interest rates adjusted for exchange rate expectations (columns (4) and

<sup>29</sup> One reason for the low share of syndicated loans on bank balance sheets is that banks sell a large fraction of the syndicated loans they originate (especially the risky ones) to institutional investors. See, for example, Irani et al. (2018).

<sup>30</sup> Table C.1 provides further robustness checks for these results, showing regressions that include firm-time and loan type fixed effects. As in Table 2, regressions are run on loan-level data.

<sup>31</sup> Specifically, we use the expected change in the USD exchange rate over the next 24 months from the Survey of Professional Forecasters.

<sup>32</sup> Uncovered interest rate parity states that  $1 + i^* = (1 + i)^f$ , where  $f$  is the expected future exchange rate expressed in units of the foreign currency per domestic currency. We approximate this condition by adjusting the observed interest rate by the 24 month exchange rate expectations. If the expected change in the exchange rate over the lifetime of the loan were different from this survey result, it would affect the estimated coefficient on the foreign currency dummy.

**Table 2**

Differences in interest rates between USD and local currency loans.

|                 | Fix rate               |                        |                        |                        | Floating rate          |                        |                        |                        |
|-----------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                 | (1)<br>Int             | (2)<br>Int             | (3)<br>UIP             | (4)<br>UIP             | (5)<br>Int             | (6)<br>Int             | (7)<br>UIP             | (8)<br>UIP             |
| FC              | −1.397***<br>(0.0956)  | −2.523***<br>(0.189)   | −0.594***<br>(0.215)   | −1.232***<br>(0.400)   | −1.019***<br>(0.131)   | −3.407***<br>(0.178)   | −0.470**<br>(0.196)    | −2.415***<br>(0.515)   |
| FC × Low Volat. |                        | 2.441***<br>(0.233)    |                        | 1.261***<br>(0.476)    |                        | 3.153***<br>(0.184)    |                        | 2.423***<br>(0.532)    |
| Ln(loan size)   | −0.0946***<br>(0.0259) | −0.0887***<br>(0.0255) | −0.0858***<br>(0.0262) | −0.0806***<br>(0.0254) | −0.0969***<br>(0.0140) | −0.0860***<br>(0.0141) | −0.0676***<br>(0.0197) | −0.0558***<br>(0.0189) |
| Ln(maturity)    | 0.0962**<br>(0.0383)   | 0.121***<br>(0.0354)   | 0.0972**<br>(0.0420)   | 0.115***<br>(0.0400)   | 0.243***<br>(0.0314)   | 0.196***<br>(0.0288)   | 0.208***<br>(0.0323)   | 0.188***<br>(0.0301)   |
| Ct-time FE      | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Rating FE       | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations    | 11,465                 | 11,062                 | 6562                   | 6560                   | 11,632                 | 11,005                 | 7455                   | 7453                   |
| R <sup>2</sup>  | 0.427                  | 0.449                  | 0.583                  | 0.588                  | 0.454                  | 0.526                  | 0.589                  | 0.607                  |

The table explores differences in interest rates borrowers pay between local currency and foreign currency loans. The sample includes only newly originated loans. In columns 1, 2, 5, and 6, the interest rate paid on a loan during the quarter of loan origination is regressed on a dummy variable *FC*, which takes the value of 1 if the loan is denominated in a currency other than the currency of the country where the borrower is located. In columns 3, 4, 7, and 8, the interest rate was adjusted for the expected change in the exchange rate over the next 24 months from the Survey of Professional forecasters. As a result, the coefficient on the variable *FC* in these columns measures the average UIP deviation in the data. *Low Volat.* is a dummy variable that takes the value of 1 for borrowers in countries with below median exchange rate volatility and zero otherwise. Columns 1 through 4 include loans with fixed interest rates, while columns 5 through 8 include floating rate loans. Standard errors are clustered by bank-quarter. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% level.

(8)). For loans to firms in countries with below median exchange rate volatilities, we do not find significant differences in interest rates between dollar and local currency loans.

### 3.2.4. The majority of borrowing is through non-syndicated loans

As Table A1 shows, 82 percent of firm-quarter observations are for firms that only borrow through non-syndicated loans, with their average borrowing being less than half of that of firms that borrow only through syndicated loans. A very small fraction of firm-quarter observations (less than 2 percent) is for firms that borrow both through syndicated and non-syndicated loans; these borrow the most. The share of foreign currency loans is fairly similar for borrowers with and without syndicated loans (62 percent versus 57 percent). For the small set of borrowers with both types of loans, foreign currency loans are more prevalent (75 percent).

### 3.2.5. The dataset covers loans in a variety of countries and industries

The 79 countries in the sample cover all world regions, as shown in Table A2. The largest share of firm-borrower observations is for high-income OECD countries (42 percent), followed by countries in Latin America and the Caribbean (24 percent). Borrowing in foreign currency is particularly prevalent in Europe and Central Asia, Latin America and the Caribbean, and Sub-Saharan Africa. Firms in all regions become past due over the sample period, but this event is more frequent for borrowers in OECD countries and Latin America and the Caribbean.

Table A3 displays loan characteristics by industry. Manufacturing firms represent 34 percent of observations. The finance and insurance industry accounts for 17 percent of observations. Loans to the finance and insurance sector are larger, have a lower risk assigned by banks, have lower interest rates, and are more often in foreign currency than loans to other sectors. The event that a borrower becomes past due on its loan payments is more frequent for the finance and insurance sector and other services than for the manufacturing, transportation, and wholesale/retail sectors.

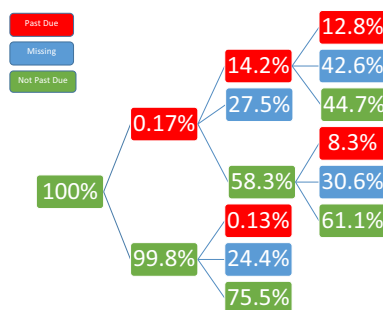
## 3.3. Past-due loans

### 3.3.1. Payment delays are a rare event

In the Y-14 data, banks report if a borrower is late on interest or principal payments.<sup>33</sup> Based on this information, we construct a variable that takes the value of 1 if a borrower becomes newly past due on her loan payments in a given quarter and zero otherwise. The event that a borrower misses a loan or principal payment after paying in the previous quarter is rare, occurring a total of 331 times in our dataset (0.23 percent of observations). That said, a frequency of 0.2 percent represents a relevant risk for banks. It implies that a loan with a 5-year maturity becomes, on average, newly past due at some point in its life with a probability of 4 percent, assuming that becoming newly past due is not auto-correlated.<sup>34</sup>

<sup>33</sup> Specifically, banks have to report a borrower as past due and the exact number of days the borrower is past due if the borrower is at least 30 days late on his payments.

<sup>34</sup> This seems a reasonable approximation, as only 30 firms are newly past due more than once in contrast to 269 firms that become past due once.



**Fig. 1.** Tracking loans' past due status over time. This figure illustrates the past-due status of loans over time. All loans start out as performing, but after one quarter ( $t + 1$ ), 0.17 percent of the loans are past due, while 99.8 percent remain performing. Of the 0.17 percent loans that become past due, 58.3 percent become performing again in  $t + 2$ , whereas 14.2 percent remain past due. 27.5 percent disappear from the sample. 12.8 percent of the 14.2 percent of loans that were past due in quarter  $t + 1$  and  $t + 2$  are still past due in  $t + 3$ .

While borrowers in emerging economies represent 49 percent of the total loan value, they account for 53 percent of newly past due borrowers.<sup>35</sup> Fig. D.2 shows the share of borrowers that become newly past due by their lagged risk rating. Among the highest rated borrowers (rating of AAA), none become past due on their loans. The share of borrowers in lower rating buckets that become past due lies between 0.1 and 0.3 percent. Fig. D.3 shows the shares of USD and local currency loans that become past due in each quarter, which vary significantly over time and across currency denominations. The chart also plots the broad dollar, a trade-weighted dollar index, from 2014q4 to 2018q4, illustrating how the dollar appreciated notably over this period.

Tables 1 and A1 show the share of observations that become past due by currency and participation type, respectively. Of note, even though local currency loans had slightly higher probabilities of default, which reflect the banks' ex-ante assessments of their riskiness, these loans became past due less often than foreign currency loans, i.e. they were ex-post less risky over the sample period. The fact that foreign currency loans turned out to be riskier than anticipated by banks may reflect the large appreciation of the dollar over that period, which was likely not fully anticipated. Comparing loans that become past due with performing loans, the former are smaller in size and have a shorter maturity, with an average size of 16 million and an average maturity of 4.8 years.

### 3.3.2. Liquidity versus solvency

Missing a loan or principal payment for 30 days or more on a large loan (i.e. becoming newly past due) indicates severe financial stress for the borrower. A key question is whether this incidence reflects a temporary liquidity problem or is the first sign of a solvency problem, that is, a default. Fig. 1 shows what happens to borrowers after they become past due. Of the 0.2 percent of borrowers that become past due, 58 percent pay in the next period. 14 percent continue to have loans that are past due, whereas 27 percent of borrowers exit the sample. The exit rate is slightly higher for borrowers that have past due loans than those that have not missed payments (24 percent). The increase in the drop-out rate likely occurs because banks write off the loans some time after they become past due, which makes them disappear from the loan book.

To obtain an estimate of the share of borrowers that default on their loans after becoming past due, we make the following assumptions. (i) A borrower is in default if she is still past due on her loans after two additional quarters. (ii) Differences in exit rates between borrowers that are past due and those that are not past due are explained by loan write-offs due to default. Based on these assumptions and the numbers shown in Fig. 1, we calculate that around 16 percent of loans that become newly past due ultimately end up in default. Our past due variable therefore captures severe liquidity problems but also reflects solvency problems for a subset of borrowers.

## 4. Empirical analysis

Proceeding in four steps, this section presents empirical evidence on the effect of exchange rate changes on firms' ability to service debt and on the determinants of firms' currency choices. First, we detail our baseline specification and show graphical evidence for the key finding. Second, we discuss the main regression results. Third, we present additional results with an emphasis on the factors that drive currency choice. Finally, we provide an extensive set of robustness checks.

<sup>35</sup> Total loan values correspond to total utilized exposures.

#### 4.1. Main empirical specification

Our main specification tests if a firm that borrows in foreign currency is more likely to default on its loan when the local currency depreciates than a firm that borrows in local currency (testable prediction 1). We employ a difference-in-difference type approach, running the following probit regression:

$$\text{past due new}_{it} = \beta_1 \Delta \ln(XR_{ct}) + \beta_2 \Delta \ln(XR_{ct}) \times FC_{it} + \beta_3 FC_{it} + \Gamma X_{it} + \alpha_{c(kr)t} + \varepsilon_{it}, \quad (16)$$

where  $\text{past due new}_{it}$  takes the value of 1 if a firm becomes past due on its loans from quarter  $t - 1$  to quarter  $t$ .<sup>36</sup>  $FC_{it}$  is the share of loans of firm  $i$  that are denominated in a foreign currency.  $\Delta \ln(XR_{ct})$  is the quarterly log change in the average bilateral exchange rate between the USD and the local currency of country  $c$  where borrower  $i$  is located (defined as local currency per USD). The key coefficient of interest  $\beta_2$  associated with the interaction term between the change in the dollar exchange rate and the foreign currency share of loans is predicted to be positive.  $X_{it}$  represents firm-level control variables (maturity, total loan volume, rating).

We are also interested in the longer-term effects of exchange rate movements. For this purpose, we run additional regressions that include both the contemporaneous quarterly change in the exchange rate as well as the cumulative exchange rate change from the date of loan origination to the preceding quarter and interactions of these two variables with the foreign currency share  $FC_{it}$ .<sup>37</sup>

The main independent variable, the change in the exchange rate, is measured as the difference between the average exchange rate in the previous quarter and the average exchange rate in the current quarter. Suppose, for example, that the local exchange rate depreciates notably in the first month of a quarter. This local depreciation will worsen the financial position and repayment ability of a firm that borrowed in foreign currency. The worsened financial position will make it less likely that the firm will still be paying its interest or principal in the next two months of the quarter. For this reason, a change in the exchange rate within a quarter can still affect the past due status within that quarter.

In all regressions, standard errors are clustered at the country-quarter level unless noted otherwise. Moreover, regressions include an extensive set of fixed effects. The most demanding specification includes country-quarter-industry-rating fixed effects. This means that we compare the effect of exchange rate changes on the probability of becoming past due between firms with foreign currency and those with local currency loans within the same country, industry, rating category, and quarter. As a result, other channels through which the exchange rate may affect firm performance are controlled for to the extent that they affect firms in the same country, industry and with the same rating in the same way, for example, through effects on demand. This also addresses concerns that results could be driven by differences in the ex-ante riskiness between firms that borrow in local currency and those that borrow in foreign currency.<sup>38</sup>

Two firms in the same country, industry, rating bucket, and quarter might still have different exchange rate exposures, for example, because they generate different shares of their incomes in foreign currency. While we cannot directly control for this selection because we do not observe firms' exposures, Corollary 1 implies that sorting generates a downward bias for our coefficient of interest  $\beta_2$ . Intuitively, a depreciation of the local currency should have a stronger negative effect on the repayment probability of a foreign currency loan for a firm that has a revealed preference for local currency debt (with a smaller foreign income stream) than for a firm that actually chooses to borrow in foreign currency (with a larger foreign income stream). Therefore, our estimate of  $\beta_2$  should represent a lower bound for the average effect of exchange rate changes on a firm's probability of becoming past due on a loan under random currency assignment.

In addition, measurement error may bias our estimate of  $\beta_2$  downward because whether a firm borrows in local or foreign currency from a U.S. bank is only a proxy for the firm's overall currency composition of debt. Generally, it would be desirable to complement our dataset with firm balance sheet data. Unfortunately, much information related to firm balance sheets is missing for foreign firms in the Y-14 data. In addition, we cannot obtain balance sheet data for a large enough sample of firms from external sources. This is because the dataset includes loans to many firms that are not publicly traded and, more importantly, the event that a borrower becomes newly past due on her loans is concentrated in smaller loans, meaning smaller firms, for which balance sheet data are not available.

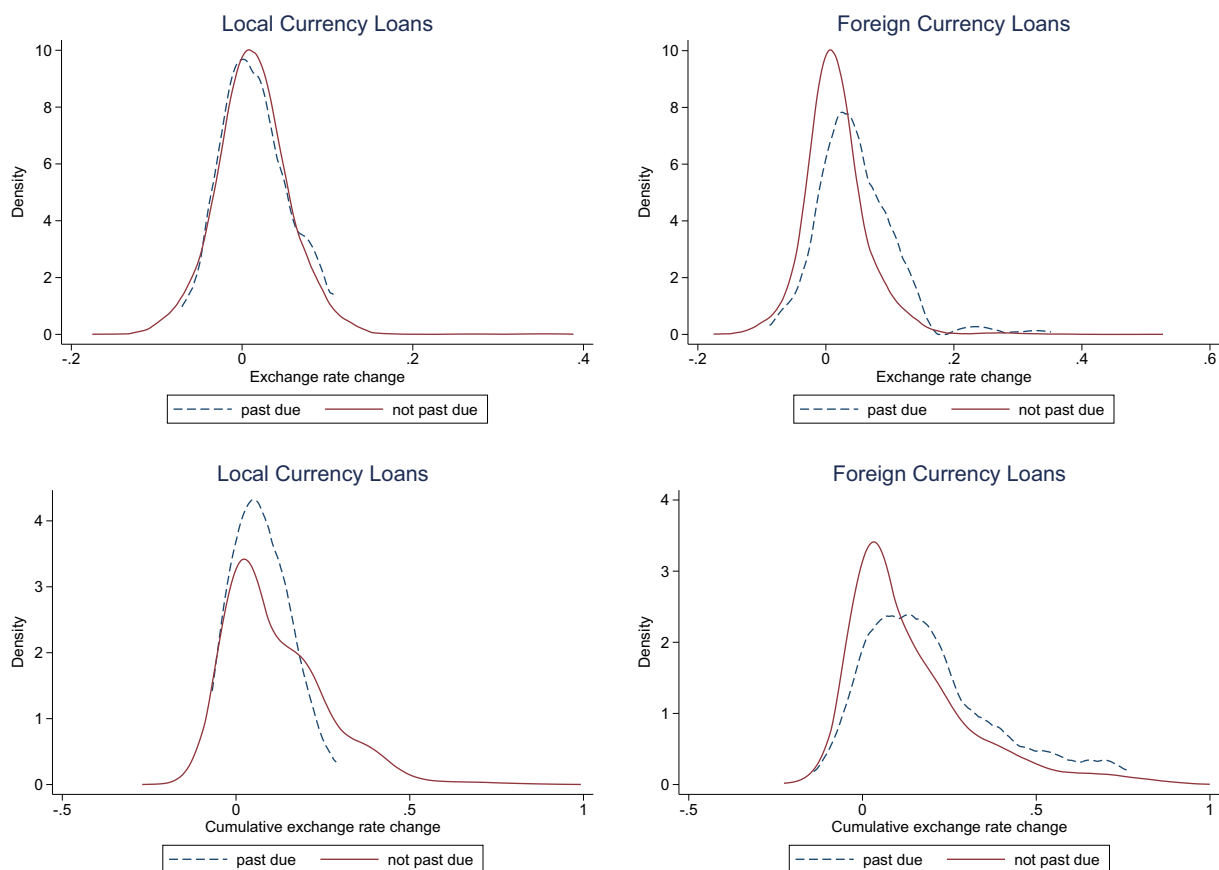
#### 4.2. Graphical evidence

Fig. 2 presents graphical evidence for our main result. Consider the two top panels that show the kernel density estimates of quarterly bilateral exchange rate changes for firms that become past due on their loans and, separately, for firms that do not become past due. The left panel has the kernel densities for firms with local currency loans, while the right panel presents those for firms with foreign currency loans. While there is no difference in the distribution of exchange rate changes between firms with local currency loans that become past due and firms with local currency loans that do not become past due (left panel), the right panel shows significant differences when the loans are denominated in foreign currency. Specifically, the distribution of

<sup>36</sup> While the model outcome is an outright default, in the data, as discussed in section 3.3.2, we observe if a firm is past due on its loan payment.

<sup>37</sup> We also ran regressions that employ the lagged change in the exchange rate as independent variable. The coefficient on the interaction term with the one-quarter lagged exchange rate has the right sign but is never significant. In contrast, as we show in section 4.3, the interaction term with the cumulative exchange rate change from loan origination to the preceding quarter is highly significant. This lagged cumulative exchange rate works better because it captures information on a larger set of exchange rate moves. At the same time, it helps address any remaining endogeneity concerns that one may have about the contemporaneous exchange rate.

<sup>38</sup> Bacchetta and Merrouche (2011) find that especially risky European borrowers switched to borrowing from U.S. banks during the European sovereign debt crisis.



**Fig. 2.** Exchange rate changes and past due status. This figure presents key findings of this paper in graphical form. The four panels show the kernel density estimates of quarterly exchange rate changes for firms that become past due on loans (dashed line) and firms that do not become past due (solid line). The left panels show kernel density estimates for firms that borrow in local currency. The top panel has quarterly exchange rate changes on the x-axis. The bottom panel shows the distribution of exchange rate changes since the origination of the loan. Exchange rates are defined as LC/USD so that a positive change indicates an appreciation of the USD against the currency of the country where the borrower resides.

firm-quarter observations that become past due is significantly shifted to the right, that is, towards larger dollar appreciations. This indicates that the more the dollar appreciates against the local currency within a quarter, the higher is the probability that a foreign currency borrower becomes newly past due. The regression results in the next section establish this result formally.

The bottom panels of Fig. 2 are constructed in parallel to the top panels but show the distribution of exchange rate changes since the loan was originated. Similar to the right top panel, the distribution of firm-quarter observations that become past due is shifted to the right, that is, towards larger dollar appreciations. Interestingly, differences are also apparent for firms that borrow in local currency. The distribution of exchange rate changes since origination for firms that do not become past due on their loans is skewed to the right relative to that of firms that become past due. This right-shift indicates that the more the dollar appreciates against the local currency since origination, the lower is the probability that a local currency borrower becomes newly past due. This finding might be reflective of the Mundell-Fleming expenditure switching channel through which firms benefit from local currency depreciation through stronger demand for domestic goods. This positive effect of local currency depreciation likely takes longer to unfold and may therefore only be evident when exchange rate changes over a longer time period are considered. Regression results shown in the next section support this interpretation.

#### 4.3. Main results

##### 4.3.1. The effect of exchange rate changes on the past-due status of loans

Table 3 displays regression results that test prediction 1. Columns (2) to (7) present the results for the specification stated in Eq. (16) with different combinations of fixed effects and control variables. Before turning to these, column (1) presents a benchmark regression without the interaction term between exchange rate changes and the foreign currency dummy. It finds that the average effect of a dollar appreciation on the past due status of loans is positive in a regression that only includes time fixed effects.

**Table 3**

Baseline results for quarterly exchange rate changes.

|                          | (1)                    | (2)                    | (3)                    | (4)                    | (5)                    | (6)                   | (7)                  |
|--------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|-----------------------|----------------------|
| D ln(XR)                 | 1.899***<br>(0.412)    | −0.431<br>(1.086)      | −1.785<br>(1.089)      |                        |                        |                       |                      |
| D ln(XR) × FC            |                        | 2.691**<br>(1.161)     | 3.479***<br>(1.161)    | 4.588***<br>(1.745)    | 4.730***<br>(1.680)    | 7.309***<br>(2.329)   | 8.023***<br>(3.112)  |
| Lagged rating            | 0.0492**<br>(0.0226)   | 0.0477**<br>(0.0226)   | 0.0454**<br>(0.0221)   | 0.0492**<br>(0.0249)   | 0.00972<br>(0.0279)    | 0.0289<br>(0.0409)    |                      |
| Ln(loan size)            | −0.0430***<br>(0.0140) | −0.0524***<br>(0.0143) | −0.0717***<br>(0.0161) | −0.0785***<br>(0.0184) | −0.0710***<br>(0.0220) | −0.105***<br>(0.0351) | −0.0622<br>(0.0434)  |
| Ln(maturity)             | −0.0467**<br>(0.0225)  | −0.0471**<br>(0.0225)  | −0.0698**<br>(0.0272)  | −0.0724**<br>(0.0308)  | −0.0853**<br>(0.0335)  | −0.0982**<br>(0.0470) | −0.129**<br>(0.0545) |
| FC                       |                        | 0.0848<br>(0.0593)     | 0.119*<br>(0.0698)     | 0.102<br>(0.0845)      | 0.0972<br>(0.0825)     | −0.0350<br>(0.113)    | −0.102<br>(0.152)    |
| D ln(XR) × Ln(loan size) |                        |                        |                        |                        | −0.205<br>(0.332)      | −0.0467<br>(0.556)    | −0.294<br>(0.710)    |
| D ln(XR) × Ln(maturity)  |                        |                        |                        |                        | 0.454<br>(0.603)       | 0.933<br>(0.757)      | 1.696<br>(1.097)     |
| D ln(XR) × L rating      |                        |                        |                        |                        | 1.291***<br>(0.434)    | 1.657***<br>(0.641)   |                      |
| Time FE                  | Yes                    | Yes                    | Yes                    | No                     | No                     | No                    | No                   |
| Ct FE                    | No                     | No                     | Yes                    | No                     | No                     | No                    | No                   |
| Ct-time FE               | No                     | No                     | No                     | Yes                    | Yes                    | No                    | No                   |
| Ct-Time-Ind FE           | No                     | No                     | No                     | No                     | No                     | Yes                   | No                   |
| Ct-Time-Ind-Rat FE       | No                     | No                     | No                     | No                     | No                     | No                    | Yes                  |
| Observations             | 147,103                | 147,103                | 119,767                | 51,811                 | 51,811                 | 9987                  | 5220                 |
| Pseudo R <sup>2</sup>    | 0.039                  | 0.045                  | 0.096                  | 0.078                  | 0.080                  | 0.143                 | 0.171                |

This table shows the effect of exchange rate changes on the past due status of borrowers. The dependent variable takes a value of 1 if a borrower becomes past due on her loans in period  $t$ .  $D \ln(XR)$  is the change in the log exchange rate between the USD and the currency of the country where the borrower resides from quarter  $t - 1$  to  $t$ . The exchange rate is defined as LC/USD. FC is the foreign currency share of a firms' loans. Standard errors are clustered by country-quarter. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% level.

Column (2) shows that exchange rate changes affect firms with foreign currency loans and those with local currency loans differently. When the interaction term between the foreign currency dummy and the exchange rate change is included in the regression, the coefficient associated with the exchange rate change ( $\beta_1$ ) turns negative and insignificant. At the same time, the coefficient of the interaction term ( $\beta_2$ ) is positive and highly significant, in line with prediction 1 and confirming the graphical evidence in the top panels of Fig. 2. The interaction term remains positive and highly significant as various types of fixed effects are included (columns (3) and (4)). Columns (5) and (6) add interaction terms between observable loan characteristics (total loan volume, average maturity, obligor rating) and the exchange rate change. As one might expect, exchange rate changes have stronger effects on the past due status of loans of lower-rated firms. Column (7) shows results for the most restrictive set of fixed effects, namely country-time-industry-rating-fixed effects. Even in this specification, the interaction term between exchange rate changes and the foreign currency dummy remains highly significant.

In Table 4, we account for exchange rate changes since origination. To disentangle the effects of contemporaneous and past exchange rate changes, we split exchange rate changes since origination into (i) the exchange rate change in the last quarter (included as a regressor in Table 3) and (ii) exchange rate changes in prior quarters after origination. The results indicate that both contemporaneous as well as past exchange rate changes play a role for the probability that a firm becomes past due on its loans. In column (3), the coefficients associated with the quarterly exchange rate change and the lagged cumulative exchange rate change are both negative and significant at standard significance levels, which suggests that dollar appreciation since loan origination decreases the probability that a firm becomes past due on its loans when it borrows in local currency. This is consistent with the expenditure switching channel through which local currency depreciation benefits firms by increasing the demand for domestic goods. The estimated coefficients associated with the interaction term between the (contemporaneous) quarterly exchange rate change and the foreign currency dummy remain positive and quantitatively very similar to those in Table 4. The interaction term between the lagged cumulative exchange rate change and the foreign currency dummy is also positive: Dollar appreciation in prior quarters (after loan origination) increases the probability that a firm with foreign currency loans becomes past due on its loans relative to a firm with local currency loans. The negative effects of foreign currency appreciation on a firm's ability to pay for its foreign currency debt are thus quite persistent and affect outcomes over the life of a loan.

#### 4.3.2. Economic significance

To understand the economic significance of the results, we compute the predicted probability that a loan becomes past due for different values of the independent variables based on the regression results shown in column (3) of Table 4. The left panel in Fig. 3 shows the predicted probability that a borrower becomes past due on its loans on the y-axis for different values of the exchange



**Table 4**

Baseline results for quarterly and lagged cumulative exchange rate changes.

|                       | (1)                 | (2)                 | (3)                  | (4)                  | (5)                 | (6)                 | (7)                |
|-----------------------|---------------------|---------------------|----------------------|----------------------|---------------------|---------------------|--------------------|
| D ln(XR)              | 1.666***<br>(0.496) | −1.484<br>(1.292)   | −2.842**<br>(1.174)  |                      |                     |                     |                    |
| D ln(XR) × FC         |                     | 3.573***<br>(1.353) | 4.420***<br>(1.240)  | 5.381***<br>(1.973)  | 5.586***<br>(1.959) | 9.359***<br>(3.059) | 11.19**<br>(4.478) |
| L Cum. D ln(XR)       | 0.159<br>(0.136)    | −0.611**<br>(0.261) | −1.188***<br>(0.313) | −1.074***<br>(0.364) | −4.955**<br>(2.404) | −5.826<br>(3.865)   | −4.131<br>(6.267)  |
| L Cum. D ln(XR) × FC  |                     | 0.832***<br>(0.280) | 1.300***<br>(0.320)  | 1.465***<br>(0.366)  | 1.281***<br>(0.389) | 1.551***<br>(0.587) | 1.720**<br>(0.801) |
| FC                    |                     | 0.0145<br>(0.0701)  | −0.0202<br>(0.0792)  | −0.0361<br>(0.0960)  | −0.0149<br>(0.0970) | −0.159<br>(0.149)   | −0.233<br>(0.226)  |
| Time FE               | Yes                 | Yes                 | Yes                  | No                   | No                  | No                  | No                 |
| Ct FE                 | No                  | No                  | Yes                  | No                   | No                  | No                  | No                 |
| Ct-time FE            | No                  | No                  | No                   | Yes                  | Yes                 | No                  | No                 |
| Ct-Time-Ind FE        | No                  | No                  | No                   | No                   | No                  | Yes                 | No                 |
| Ct-Time-Ind-Rat FE    | No                  | No                  | No                   | No                   | No                  | No                  | Yes                |
| Observations          | 99,981              | 99,981              | 72,009               | 32,711               | 32,711              | 5611                | 2994               |
| Pseudo R <sup>2</sup> | 0.038               | 0.048               | 0.090                | 0.090                | 0.097               | 0.164               | 0.194              |

This table shows how changes in the exchange rate from the date of origination of the loan to the prior quarter affect the past due status of loans in addition to effects of contemporaneous changes in the exchange rate. The dependent variable takes the value of 1 if a borrower becomes past due on her loans in period  $t$ .  $D \ln(XR)$  is the change in the log exchange rate between the USD and the currency of the country where the borrower resides from quarter  $t - 1$  to  $t$ .  $L \text{ Cum. } D \ln(XR)$  is the change in the log exchange rate from the date when the loan was originated to quarter  $t - 1$ . The exchange rate is defined as LC/USD. FC is the foreign currency share of a firms' loans. The variables log maturity, log loan size, and the lagged rating as well as interaction terms between these variables and the exchange rate changes are included as controls but are not displayed. Standard errors are clustered by country-quarter. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% level.

rate change ( $\Delta \ln(XR_{ct})$ ) on the x-axis. 95 percent confidence intervals around the predicted probabilities are indicated by the vertical lines. The dashed line is for borrowers that only borrow in local currency. The solid line is for borrowers that borrow in foreign currency. The chart confirms that exchange rate changes have a differential effect on the past due status of borrowers depending on whether they borrow in local or foreign currency with stronger effects for borrowers with foreign currency loans. When the dollar appreciates by 10 percent, the predicted probability of becoming past due is 0.09 percent for local currency borrowers. In contrast, the predicted probability is 41 basis points higher for firms that borrow in foreign currency, at 0.5 percent. The average marginal effect of the interaction term is 0.042, meaning that the marginal effect of a 10-percent appreciation of the dollar is 42 basis points higher on average for firms with foreign currency loans relative to firms with local currency loans.<sup>39</sup> Compared to a firm's baseline probability of becoming past due on its loans of 0.2 percent (the frequency of new past due status in our data), these differential effects are economically significant.

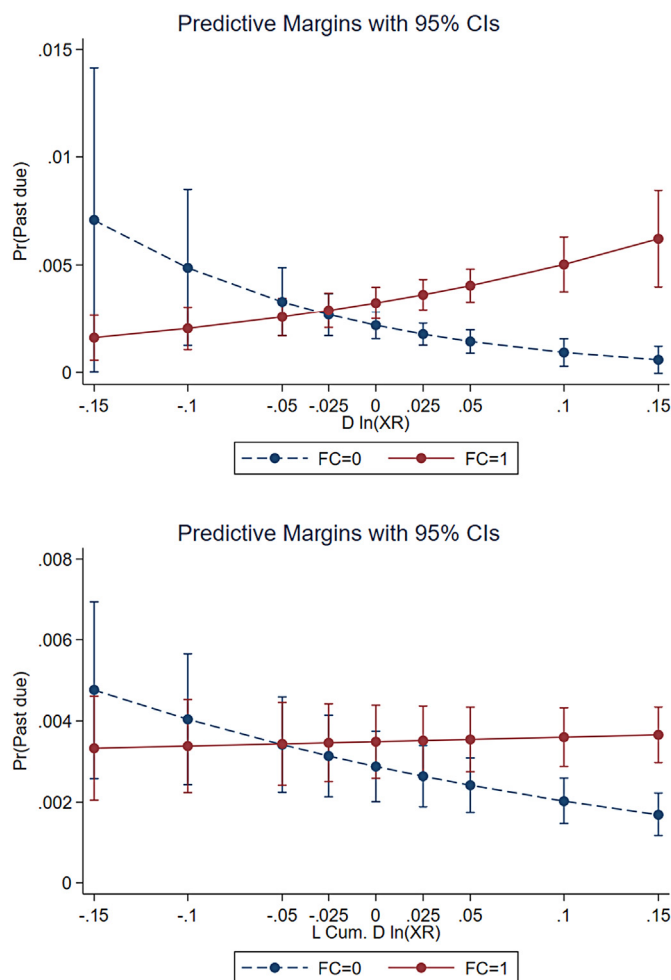
The right panel in Fig. 3 is constructed in parallel to the left panel, showing the predicted probability of becoming past due as a function of cumulative exchange rate changes up to quarter  $t - 1$ . Again, if the dollar has appreciated since the origination of the loan up to quarter  $t - 1$ , firms with foreign currency loans have a higher probability of becoming past due on their loans than local currency borrowers. The average marginal effect on the interaction term that includes the cumulative exchange rate change is 0.01. Thus the marginal effect of a 10-percent increase in the exchange rate from the quarter of origination to quarter  $t - 1$  on the past due status of loans is 10 basis points higher for borrowers with foreign currency loans than for those with local currency loans.

#### 4.3.3. Appreciations versus depreciations

As both figure indicate, the effects of exchange rate changes on the past due status of local currency borrower and foreign currency borrowers are only statistically different from each other for depreciations of the local currency (indicated by positive values on the x-axis) but not for appreciations (indicated by negative values on the x-axis). Table 5 presents regressions to study this relationship formally. The regressions shown in the table include a triple interaction between the exchange rate change, the foreign currency dummy, and a dummy variable that takes the value of 1 if the local exchange rate depreciated against the dollar and zero otherwise. Columns (1) and (2) show results for quarterly exchange rate changes. Columns (3) and (4) use exchange rate changes since loan origination instead. The coefficients associated with the triple interactions are always positive and significant. That is, the differential effect of exchange rate changes on firms with foreign currency loans relative to firms with local currency loans is stronger when the local currency depreciates. This finding is consistent with our model if we assume that on average firms that borrow in foreign currency have larger currency mismatches than firms that borrow in local currency.<sup>40</sup>

<sup>39</sup> To compute marginal effects of interaction terms in the non-linear Probit model, we follow Williams (2012) and Pinzon (2016).

<sup>40</sup> Note that this is a natural assumption as most firms (including exporters) have most of their sales at home. See, for example, Benavente et al. (2003).



**Fig. 3.** Predicted probabilities of becoming newly past due. This figure shows the predicted probability that a borrower becomes past due on his/her loans on the y-axis for different values of the exchange rate change as indicated on the x-axis. 95 percent confidence intervals around the predicted probabilities are indicated by the vertical lines. The dashed line is for borrowers that only borrow in local currency. The solid line is for borrowers that borrow in foreign currency. The top panel has the quarterly exchange rate change on the x-axis ( $\Delta \ln(XR_{it})$ ). The bottom panel shows  $\text{Cum} \Delta \ln(XR_{it-1})$ , the cumulative exchange rate change from the origination date of the loans to quarter  $t - 1$ .

#### 4.3.4. Syndicated versus regular loans

As discussed earlier, the majority of credit that the dataset captures is not in the form of syndicated loans. Table 6 shows regression results where the sample is split between borrowers without syndicated loans and those with syndicated loans.<sup>41</sup> To preserve the largest possible sample size, regressions include time fixed effects in line with column (2) of Table 4. Coefficients of interest are only significant for the sample of borrowers without syndicated loans. That is, among firms that borrow through syndicated loans, we do not find differential effects of exchange rate changes on past due status for firms with and without foreign currency loans.<sup>42</sup> These results underline the value of the Y-14 loan-level data, which also provide cross-country information for small non-syndicated loans to likely smaller, less sophisticated firms that may manage exchange rate risk less well.

#### 4.3.5. Alternative dependent variable: changes in borrowers' ratings and probabilities of default

As discussed, the event that a borrower becomes past due on her loans is quite rare. In contrast, changes in ratings and probabilities of default assigned by the banks are more frequent. To complement our baseline analysis on the past due status, we analyze two additional outcome variables that measure these more frequent events; first, a dummy variable that indicates whether a borrower was downgraded since the previous quarter.<sup>43</sup> This happens for 3 percent of the borrower-quarter observations. Second,

<sup>41</sup> We assign borrowers for whom syndicated loans constitute less than 10 percent of the total loan volume to the non-syndicated group.

<sup>42</sup> Firms that obtain syndicated loans tend to be larger, more sophisticated firms that may be better able to hedge their exchange risk. At the same time, banks tend to retain syndicated loans with better ratings, selling off large portions of the risky syndicated loans, so-called leveraged loans. As a result, we mostly observe syndicated loans to very safe borrowers. In the sample, borrowers of syndicated loans become past due only 31 times, limiting the statistical power of the exercise.

<sup>43</sup> Downgrades refer to a deterioration in the risk rating the bank's internal risk model assigns, which we observe in the Y-14 data. Most firms in the data do not have public credit ratings offered by credit agencies.

**Table 5**

Depreciation vs. appreciation.

|                       | Quart. change        |                     | Cumm. change         |                      |
|-----------------------|----------------------|---------------------|----------------------|----------------------|
|                       | (1)                  | (2)                 | (3)                  | (4)                  |
| D ln(XR)              | 9.167<br>(6.932)     |                     | 4.847*<br>(2.939)    | 4.652<br>(3.499)     |
| D ln(XR) × FC         | −8.810**<br>(4.273)  | −11.37*<br>(5.831)  | −8.491***<br>(2.962) | −8.699***<br>(3.362) |
| FC                    | −0.307**<br>(0.125)  | −0.407**<br>(0.166) | −0.320*<br>(0.164)   | −0.300<br>(0.184)    |
| D ln(XR) × depr.      | −11.59***<br>(3.926) |                     | −9.856***<br>(2.481) | −11.14***<br>(2.934) |
| D ln(XR) × FC × depr. | 12.20***<br>(4.719)  | 15.84**<br>(6.323)  | 11.09***<br>(3.034)  | 11.74***<br>(3.450)  |
| FC × depr.            | 0.485***<br>(0.155)  | 0.565***<br>(0.212) | 0.186<br>(0.167)     | 0.119<br>(0.191)     |
| Deprec.               | −0.333***<br>(0.125) |                     | −0.182<br>(0.127)    | −0.176<br>(0.151)    |
| Ct FE                 | Yes                  | No                  | Yes                  | No                   |
| Time FE               | Yes                  | No                  | Yes                  | No                   |
| Ct-time FE            | No                   | Yes                 | No                   | Yes                  |
| Observations          | 119,767              | 51,811              | 96,537               | 45,257               |
| Pseudo R <sup>2</sup> | 0.102                | 0.084               | 0.102                | 0.096                |

This table analyzes potential asymmetric effects of currency appreciations and depreciations. The dependent variable takes the value of 1 if a borrower becomes past due on her loans in period  $t$ . In columns 1 and 2,  $D \ln(XR)$  stands for the change in the log exchange rate between the USD and the currency of the country where the borrower resides from quarter  $t - 1$  to quarter  $t$ . In columns 3 and 4, it stands for the change in the log exchange rate from the date when the loan was originated to quarter  $t$ . The exchange rate is defined as LC/USD. FC is the foreign currency share of a firms' loans. *depr.* represents a dummy variable that takes the value of 1 if  $D \ln(XR) > 0$ , meaning that the local currency depreciated. The variables log maturity, log loan size, and the lagged rating as well as interaction terms between these variables and the exchange rate changes are included as controls but are not displayed. Standard errors are clustered by country-quarter. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% level.

**Table 6**

Syndicated loans versus non-syndicated loans.

|                       | Non-syndicated     |                     | Syndicated         |                      |
|-----------------------|--------------------|---------------------|--------------------|----------------------|
|                       | (1)                | (2)                 | (3)                | (4)                  |
|                       | Quart.             | Cumm.               | Quart.             | Cumm.                |
| D ln(XR)              | 5.835<br>(4.416)   | 1.191<br>(1.662)    | −21.49<br>(18.76)  | −14.44***<br>(3.456) |
| D ln(XR) × FC         | 2.997**<br>(1.186) | 1.532***<br>(0.318) | 0.816<br>(2.453)   | −0.560<br>(0.578)    |
| FC                    | 0.0824<br>(0.0632) | −0.0102<br>(0.0676) | 0.00479<br>(0.126) | 0.0651<br>(0.145)    |
| Time FE               | Yes                | Yes                 | Yes                | Yes                  |
| Observations          | 120,988            | 102,187             | 19,744             | 18,973               |
| Pseudo R <sup>2</sup> | 0.049              | 0.052               | 0.034              | 0.044                |

This table presents results separately for borrowers with (columns 1–2) and without syndicated loans (columns 3–4). A borrower is assigned to the non-syndicated loans bin if less than 10 percent of the borrower's total loan amount (in terms of utilized exposure) is associated with syndicated loans. The dependent variable takes the value of 1 if a borrower becomes past due on her loans in period  $t$ . In columns 1 and 3,  $D \ln(XR)$  stands for the change in the log exchange rate between the USD and the currency of the country where the borrower resides from quarter  $t - 1$  to quarter  $t - 1$ . In columns 2 and 4 instead, it stands for the change in the log exchange rate from the date when the loan was originated to quarter  $t$ . The exchange rate is defined as LC/USD. FC is the foreign currency share of a firms' loans. The variables log maturity, log loan size, and the lagged rating as well as interaction terms between these variables and the exchange rate changes are included as controls but are not displayed. Standard errors are clustered by country-quarter. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% level.

we create a variable that takes the value of 1 if the bank increased the probability of default assigned to the borrower compared to the previous quarter, which happens for 7 percent of observations. Table 7 presents the regression results using these alternative dependent variables. The change in the exchange rate  $\Delta \ln(XR_{ct})$  in these regression is the quarterly change. We do not look at cumulative exchange rate changes here because the effects of past exchange rate changes should already be incorporated in banks' previous credit risk assessments. Consistent with the results on the past due status of loans, a firm with foreign currency loans is more likely to be downgraded or assigned a higher probability of default by the bank when the dollar appreciates than a

**Table 7**

Ratings downgrades and increases in PD.

|                       | Ratings change       |                      | PD change           |                     |
|-----------------------|----------------------|----------------------|---------------------|---------------------|
|                       | (1)                  | (2)                  | (3)                 | (4)                 |
| D ln(XR)              | −0.450<br>(0.375)    |                      | −0.0373<br>(0.423)  |                     |
| D ln(XR) × FC         | 0.877**<br>(0.444)   | 0.880**<br>(0.445)   | 0.851*<br>(0.452)   | 0.529<br>(0.420)    |
| FC                    | −0.0399*<br>(0.0211) | −0.0401*<br>(0.0213) | −0.0298<br>(0.0251) | −0.0311<br>(0.0240) |
| Time FE               | Yes                  | No                   | Yes                 | No                  |
| Ct FE                 | Yes                  | No                   | Yes                 | No                  |
| Ct-time FE            | No                   | Yes                  | No                  | Yes                 |
| Observations          | 146,357              | 134,203              | 108,586             | 104,458             |
| Pseudo R <sup>2</sup> | 0.020                | 0.039                | 0.024               | 0.051               |

This table shows results when the dependent variable is the probability that a borrower was downgraded (columns 1–2) or the probability that a borrower's probability of default increased (columns 3–4) from quarter  $t - 1$  to quarter  $t$ .  $D \ln(XR)$  stands for the change in the log exchange rate between the USD and the currency of the country where the borrower resides from quarter  $t - 1$  to quarter  $t - 1$  throughout. The exchange rate is defined as LC/USD. FC is the foreign currency share of a firms' loans. The variables log maturity, log loan size, and the lagged rating as well as interaction terms between these variables and the exchange rate changes are included as controls but are not displayed. Columns (1) and (2) do not include lagged ratings or their interactions with exchange rate changes as controls. Standard errors are clustered by country-quarter. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% level.

**Table 8**

Foreign sales as a natural hedge.

|                            | Quart. change        |                     | Cumm. change         |                      |
|----------------------------|----------------------|---------------------|----------------------|----------------------|
|                            | (1)                  | (2)                 | (3)                  | (4)                  |
| D ln(XR)                   | −15.19**<br>(7.093)  |                     | −7.155***<br>(1.947) | −7.907***<br>(2.529) |
| D ln(XR) × FC              | 5.136***<br>(1.592)  | 7.336***<br>(2.498) | 2.519***<br>(0.573)  | 3.100***<br>(0.772)  |
| FC                         | 0.243**<br>(0.102)   | 0.198<br>(0.130)    | 0.00459<br>(0.114)   | −0.0403<br>(0.132)   |
| D ln(XR) × for. sales      | 6.033***<br>(2.313)  | 7.795**<br>(3.370)  | 1.165<br>(1.079)     | 1.554<br>(1.358)     |
| D ln(XR) × FC × for. sales | −7.131**<br>(3.632)  | −9.068*<br>(4.764)  | −2.965**<br>(1.330)  | −3.512**<br>(1.649)  |
| FC × for. sales            | −0.500***<br>(0.187) | −0.474**<br>(0.230) | −0.210<br>(0.219)    | −0.181<br>(0.260)    |
| For. sales share           | −0.0273<br>(0.134)   | −0.0992<br>(0.163)  | 0.000333<br>(0.159)  | −0.0475<br>(0.195)   |
| Ct FE                      | Yes                  | No                  | Yes                  | No                   |
| Time FE                    | Yes                  | No                  | Yes                  | No                   |
| Ct-time FE                 | No                   | Yes                 | No                   | Yes                  |
| Observations               | 100,353              | 40,582              | 79,357               | 34,879               |
| R <sup>2</sup>             |                      |                     |                      |                      |
| Pseudo R <sup>2</sup>      | 0.099                | 0.083               | 0.100                | 0.093                |

This table analyzes whether the effect of exchange rate changes differs across industries with varying shares of international sales in total sales. The dependent variable takes the value of 1 if a borrower becomes past due on her loans in period  $t$ . In columns 1 and 2,  $D \ln(XR)$  stands for the change in the log exchange rate between the USD and the currency of the country where the borrower resides from quarter  $t - 1$  to quarter  $t - 1$ . In columns 3 and 4 instead, it stands for the change in the log exchange rate from the date when the loan was originated to quarter  $t$ . The exchange rate is defined as LC/USD. FC is the foreign currency share of a firms' loans. *for. sales* represents the average share of sales that go abroad for an industry-country-quarter combination derived from Worldscope. The variables log maturity, log loan size, and the lagged rating as well as interaction terms between these variables and the exchange rate changes are included as controls but are not displayed. Standard errors are clustered by country-quarter. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% level.

firm that borrows in local currency. The average marginal effect of the interaction term in column (1) is 0.06, meaning that foreign currency loan borrowers are 60 basis points more likely to be downgraded after a 10-percent currency depreciation than local currency borrowers. Similarly, foreign currency borrowers are 120 basis points more likely to be assigned a higher probability of default than local currency borrowers based on the regression results shown in column (3).

#### 4.4. Additional results

##### 4.4.1. Foreign sales as a natural hedge

Prediction 2 states that the effect of a depreciation of the local currency on the default probability of firms with foreign currency loans relative to that of firms with local currency loans is smaller when firms have a natural hedge, that is, when they face a smaller currency mismatch because of income streams in the currency of the loan. While we do not observe foreign currency income at the firm level, we have available information on income generated abroad by industry, which can shed some light on the empirical validity of the prediction. Table 8 shows results from a specification that includes a triple interaction between the exchange rate change, the foreign currency dummy, and the share of foreign sales in total sales specific to a quarter, industry and country.<sup>44</sup>

Columns (1) and (2) present results for quarterly exchange rate changes, whereas columns (3) and (4) show results for exchange rate changes since loan origination. In all columns, the coefficients associated with the triple interactions are negative and highly significant. Thus, the differential effect of exchange rate changes on firms that borrow in foreign currency relative to firms that borrow in local currency is smaller the larger the share of foreign sales in total sales is. In other words, when firms belong to industries with a larger share of foreign sales in total sales, this reduces the effects of exchange rate changes on loan payments, suggesting that foreign income may serve as a natural hedge for firms. We calculate that, on average, there is no difference in the effect of quarterly exchange rate changes between firms that borrow in local currency and firms that borrow in foreign currency when they operate in an industry where roughly 60 percent of sales go to foreign countries. The predicted probability that a firm becomes past due given a 10-percent depreciation of the local currency is 45 basis points higher for a firm with foreign currency loans relative to a firm with local currency loans when the firms are in industries with a foreign sales share in the 25th percentile of the distribution (12 percent foreign sales). This compares to a 5-basis-point difference for firms in the 75th percentile of the foreign sales share distribution (56 percent foreign sales).

##### 4.4.2. Selection into borrowing in foreign currency

The theoretical model predicts that firms with foreign currency income are more likely to choose to borrow in foreign currency (prediction 3). At the same time, more firms should borrow in foreign currency if the foreign risk-free rate is below the domestic risk-free rate (prediction 4). Table 9 explores these predictions. In column (1), a dummy variable that takes the value of 1 if the loan is denominated in foreign currency is regressed on the country-industry-time varying share of foreign sales, the country-time varying difference in average interest rates between a local currency and a USD loan, as well as several loan characteristics (log loan size, and log maturity). Associated marginal effects from the probit regression are shown in column (2). In column (3), the simple difference in average interest rates is replaced by the average UIP deviation, where interest rates have been adjusted for expected exchange rate changes following the approach described in Section 3.2.<sup>45</sup> Columns (4) and (5) repeat the regressions with different combinations of fixed effects. The coefficients on the foreign sales share are highly significant and positive throughout suggesting that firms in industries with a higher share of foreign sales are more likely to borrow in foreign currency than firms in industries that sell more domestically. Similarly, a larger difference between the interest rate paid for the local currency loan and that paid for the USD loan increases the probability that a firm borrows in foreign currency. Moreover, foreign currency loans are larger and have shorter maturities than local currency loans.<sup>46</sup>

Quantitatively, interest rate differentials are a more important determinant of currency choice than the industry-level share of foreign sales. When the interest rate differential is increased by 0.029, which corresponds to going from the 25th to the 75th percentile of the distribution, the probability that a firm chooses the foreign currency loan increases, on average, by 150 percent. In comparison, an increase in the industry-level foreign sales share by 45 percentage points (increase from the 25th percentile to the 75th percentile) increases the probability that a firm chooses the foreign currency loan by 5 percent on average.

#### 4.5. Robustness

This section presents several robustness exercises. In particular, results are robust to accounting for differences between variable- and fixed-rate loans. We also show that results are not driven by changes in the oil price, which is highly correlated with changes in the dollar exchange rate over the sample period. Additional checks indicate that results are not driven by individual countries, the financial sector, or borrowers in countries experiencing domestic crises.

##### 4.5.1. Alternative estimation methods

Table C.2 shows the regression results when alternative estimation methods are employed to estimate the same specification as in column (3) of Table 4. The coefficient on the interaction term with  $\Delta(XR_{ct})$  obtained from OLS estimation is 0.045 and

<sup>44</sup> Information on the foreign sales share is from Worldscope and is calculated by averaging the shares for all available firms within a country, industry and quarter.

<sup>45</sup> The interest rate differential is calculated as the difference between the average interest rate of a local currency loan and that of a foreign currency loan by country and quarter. As an alternative, we first regressed interest rates on log size, log maturity, and rating and used the residuals of this regression to calculate the average interest rate differential. Results are basically identical.

<sup>46</sup> The literature has uncovered several firm characteristics that play a role in foreign currency borrowing. As shown by Gelos (2003) and Aguiar (2005), size plays a role with larger firms being more likely to borrow in a foreign currency. Allayannis et al. (2003) and Mora et al. (2013) find that less opaque firms with easily verifiable collateral, higher net worth, and greater tangible assets have more dollar debt. The evidence on the role of profits and leverage for the currency denomination of loans is mixed. See also Brown and De Haas (2012b).

**Table 9**

Determinants of the denomination of a loan.

|                     | (1)<br>Reg. coeff.    | (2)<br>Margin. eff.     | (3)<br>Reg coeff.     | (4)<br>Reg coeff.     | (5)<br>Reg coeff.     |
|---------------------|-----------------------|-------------------------|-----------------------|-----------------------|-----------------------|
| Foreign Sales Share | 0.147**<br>(0.0682)   | 0.0478**<br>(0.0221)    | 0.335***<br>(0.0790)  | 0.163***<br>(0.0631)  | 0.405***<br>(0.0826)  |
| Ln(loan size)       | 0.101***<br>(0.0101)  | 0.0326***<br>(0.00323)  | 0.113***<br>(0.0121)  | 0.107***<br>(0.0101)  | 0.0903***<br>(0.0102) |
| Ln(maturity)        | −0.165***<br>(0.0182) | −0.0533***<br>(0.00575) | −0.118***<br>(0.0210) | −0.176***<br>(0.0174) | −0.159***<br>(0.0173) |
| Int. rate. diff.    | 4.713***<br>(1.252)   | 1.528***<br>(0.406)     |                       |                       |                       |
| UIP diff.           |                       |                         | 2.649***<br>(0.940)   |                       |                       |
| Constant            | 0.637<br>(0.526)      |                         | 0.293<br>(0.591)      | 1.413*<br>(0.800)     | 0.570<br>(0.801)      |
| Time FE             | Yes                   | No                      | Yes                   | No                    | No                    |
| Country FE          | Yes                   | No                      | Yes                   | No                    | No                    |
| Rating FE           | Yes                   | Yes                     | Yes                   | Yes                   | Yes                   |
| Country-time FE     | No                    | No                      | No                    | Yes                   | Yes                   |
| Industry-time FE    | No                    | No                      | No                    | No                    | Yes                   |
| Observations        | 15,888                | 15,888                  | 11,285                | 15,948                | 15,719                |
| Pseudo $R^2$        | 0.152                 |                         | 0.143                 | 0.191                 | 0.227                 |

This table explores the factors that determine whether a loan is denominated in foreign or domestic currency. The dependent variable takes the value of 1 if the loan is denominated in a currency other than that of the country where the borrower resides. *Foreign Sales Share* represents the average share of sales that go abroad specific to an industry-country-quarter combination derived from *Worldscope*. *Int. rate. diff* stands for the difference in the average interest rate of a local currency loan and a USD loan specific to a country and quarter. *UIP diff.* represents the average UIP deviation for loans to a country in a given quarter, calculated as the difference in average interest rates where interest rates have been adjusted for exchange rate expectations. With the exception of column 2, all columns show regression results, while column 2 displays marginal effects associated with the coefficients shown in column 1. Standard errors are clustered by country-quarter. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% level.

quantitatively very similar to the computed average marginal effect of the interaction term from the probit regression. The calculated average marginal effect of the interaction term based on logit results is 0.041. Significance levels are also very similar across specifications. In appendix Table C.9, we present the results when the specifications in Table 4 are estimated using OLS.

#### 4.5.2. Controlling for the interest rate

So far, we have included as controls a firm's total loan volume, maturity of its loans, and its lagged rating, and interactions of these variables with exchange rate changes. In a next step, we also include the lagged average interest rate a firm pays on its loans directly and as an interaction with the quarterly exchange rate change. Differences in interest rates between firms may contain additional information that might be useful to control for. We allow the effect of the lagged interest rate to differ across floating-rate and fixed-rate loans, interacting the variables of interest with two dummies that indicate floating- or fixed-rate loans, respectively. Results are shown in Table C.3 and are qualitatively and quantitatively little changed when differences in interest rates are accounted for.

#### 4.5.3. Split by loan type

Borrowers with floating-rate loans and those with fixed-rate loans might behave differently in response to exchange rate changes, in particular as floating rates are directly linked to reference rates and hence macroeconomic and financial shocks. To provide further robustness, we run regressions for borrowers with fixed-rate loans and those with floating-rate loans separately. Results are presented in Table C.4. Based on columns (1) and (3), the main coefficient of interest is positive and highly significant for borrowers with fixed-rate loans and similar in magnitude to those obtained for the sample of borrowers with variable-rate loans. As an alternative to the sample splits shown in Table C.4, we include country-quarter-interest-rate-type fixed effects in the regressions in Table C.5. While the standard errors increase for the regressions that employ quarterly changes in the exchange rate (columns (1) through (3)), the coefficient estimates remain highly significant and quantitatively similar with the additional set of fixed effects when the exchange rate change since origination is used (shown in columns (4) through (6)).

#### 4.5.4. Additional checks

Table C.6 presents additional robustness checks. Column (1) is based on a sample of banks that are present in the data set throughout. Column (2) adds bank-time fixed effects. Column (3) is based on a sample that includes only borrowers in advanced economies excluding the U.S. Column (4) has only borrowers in emerging markets. In column (5), borrowers in the financial sector are dropped from the sample. Column (6) excludes observations associated with a dollar appreciation of more than 15 percent quarter-on-quarter. Column (7) excludes several countries that experienced a crisis between 2014 and 2018, namely Russia, Turkey, Brazil, and Argentina. The coefficients associated with the interaction between the exchange rate change and  $FC_{it}$  remain



significant at standard significance levels throughout with the exception of columns (3) and (4). The fact that the coefficients associated with the interaction term are not significant when the sample is split by region while the magnitudes of the coefficients are fairly similar suggests that results are driven by both advanced economy and emerging market firms.

Finally, to exclude that our results are driven by changes in the prices of oil or commodities more broadly, which were highly correlated with the dollar over our sample period, we run two additional robustness exercises. First, we include changes in the oil price in our regressions in Table C.7. Second, we exclude firms in commodity sectors from the sample in Table C.8. In both tables, regression results remain very similar to the baseline results.

## 5. Conclusions

Employing unique data on U.S. banks' loans to foreign firms, this paper explores the effect of exchange rate changes on the ability of firms to repay their debt when they borrow in foreign currency. The estimation results indicate that a depreciation of the local currency makes a firm with foreign currency debt less likely to repay its loans. Even though firms tend to borrow more in foreign currency when they belong to industries that generate more income abroad, firms remain exposed to significant exchange rate risk. The sample used in the empirical analysis covers the years 2014 to 2018, a period of generally benign economic and financial conditions. Movements in the exchange rate will likely have stronger effects on firm's past due status when the domestic economy is faltering, the supply of credit is more constrained, and exchange rate changes are more drastic, for example, during currency or banking crises.

As the paper indicates, firms' exchange rate exposure can translate into credit risk for banks, which see a larger share of foreign currency loans become past due when the dollar appreciates. For large U.S. banks, this component of credit risk is relatively limited though because foreign currency lending constitutes a small share of these banks' overall activities and because, at least in normal times, the absolute share of past-due loans is small. Also, banks may find ways to hedge against this indirect exchange rate exposure. This could be different for banks from other countries. Large U.S. banks are monitored and regulated by the Federal Reserve. In fact, the Y-14 data this paper employs are submitted for stress testing purposes. Lending practices of other banks that provide foreign currency loans may not be as closely watched and credit risk associated with borrowers' exchange rate exposures could be much more material, as past historical episodes highlight, especially when banks lend to lower quality borrowers.

A caveat of this paper is the lack of information on firms' overall exchange rate exposure. It would be highly desirable to draw on more comprehensive information on firms' debt denomination, foreign currency income and hedging practices to further assess firms' exchange rate risk. Similarly, it would be interesting to understand to which degree banks might hedge the indirect exchange rate exposure in their loan books.

## Data availability

Replication Codes for "Foreign Currency Loans and Credit Risk: Evidence from U.S. Banks" (Reference Data) (Mendeley Data)

## Appendix A. Appendix tables

**Table A1**

Loan characteristics, syndicated vs. non-syndicated.

|                              | (1)            |      | (2)        |      | (3)   |      |
|------------------------------|----------------|------|------------|------|-------|------|
|                              | Non-syndicated |      | Syndicated |      | Mixed |      |
|                              | mean           | p50  | mean       | p50  | mean  | p50  |
| Util. Exposure (USD million) | 20.9           | 5.08 | 46.8       | 26.2 | 132.2 | 74.9 |
| Maturity (years)             | 5.54           | 4.34 | 5.84       | 5.01 | 6.00  | 5.19 |
| Prob. of default (pct)       | 1.81           | 0.68 | 2.02       | 0.64 | 1.81  | 0.53 |
| Interest rate (pct)          | 3.53           | 2.75 | 2.83       | 2.61 | 2.66  | 2.30 |
| Loans in for. currency (pct) | 0.57           | 1    | 0.62       | 1    | 0.75  | 1    |
| Loans newly past due (pct)   | 0.25           | 0    | 0.10       | 0    | 0.26  | 0    |
| Observations                 | 120,873        |      | 23,506     |      | 2724  |      |

The table shows summary statistics of the baseline sample with 147,103 observations where borrowers are grouped according to whether they borrow through syndicated loans or not. Column 1 includes borrowers that do not have syndicated or participated loans. Column 2 has borrowers that only take out syndicated or participated loans. Column 3 includes borrowers that have both types of loans. The table displays the means and the medians of the following variables: utilized exposure, maturity, bank-internal probability of default, interest rate, a dummy variable which is one when the borrower becomes past due on its loans in period  $t$ , and the share of a borrower's loans that are not denominated in the currency of the country where the borrower is located.

**Table A2**

Loan characteristics, by region.

|                              | (1)    |      | (2)   |      | (3)    |      | (4)      |      | (5)    |      | (6)  |      | (7)   |      | (8)  |      |
|------------------------------|--------|------|-------|------|--------|------|----------|------|--------|------|------|------|-------|------|------|------|
|                              | EAP    |      | ECA   |      | OECD   |      | Non-OECD |      | LAC    |      | MENA |      | SA    |      | SSA  |      |
|                              | mean   | p50  | mean  | p50  | mean   | p50  | mean     | p50  | mean   | p50  | mean | p50  | mean  | p50  | mean | p50  |
| Util. Exposure (USD million) | 22.7   | 6    | 35.4  | 9.75 | 32.2   | 9.24 | 24.6     | 4.52 | 21.5   | 5.23 | 18.7 | 9    | 20.6  | 6.11 | 30.9 | 15.7 |
| Maturity (years)             | 5.52   | 4.01 | 5.69  | 4.85 | 5.72   | 5.00 | 5.83     | 4.61 | 5.05   | 4.27 | 7.92 | 5.73 | 6.10  | 5.08 | 6.41 | 5.00 |
| Prob. of default (pct)       | 1.39   | 0.67 | 2.06  | 0.81 | 1.68   | 0.64 | 2.06     | 0.71 | 2.12   | 0.71 | 2.65 | 1.08 | 2.10  | 1.04 | 2.31 | 0.66 |
| Interest rate (pct)          | 3.46   | 2.99 | 5.10  | 3.10 | 2.64   | 2.50 | 2.42     | 2.12 | 4.24   | 3.17 | 3.90 | 3.92 | 4.93  | 3.92 | 4.43 | 3.50 |
| Loans newly past due (pct)   | 0.017  | 0    | 0.049 | 0    | 0.26   | 0    | 0.034    | 0    | 0.41   | 0    | 0.22 | 0    | 0.043 | 0    | 0.15 | 0    |
| Loans in for. currency (pct) | 0.70   | 1    | 0.83  | 1    | 0.47   | 0    | 0.66     | 1    | 0.70   | 1    | 0.59 | 1    | 0.41  | 0    | 0.76 | 1    |
| Observations                 | 17,905 |      | 6158  |      | 62,330 |      | 11,667   |      | 35,743 |      | 1367 |      | 9291  |      | 2642 |      |

The table shows summary statistics of the baseline sample with 147,103 observations grouped by region. EAP: East Asia & Pacific, ECA: Europe & Central Asia, OECD: High-income OECD countries, non-OECD: High-income non-OECD countries, LAC: Latin America and the Caribbean, MENA: Middle East and North Africa, SA: South Asia, SSA: Sub-Saharan Africa. The table displays the means and the medians of the following variables: utilized exposure, maturity, bank-internal probability of default, interest rate, a dummy variable which is one when the borrower becomes past due on its loans in period  $t$ , and the share of a borrower's loans that are not denominated in the currency of the country where the borrower is located.

**Table A3**

Loan characteristics, by sector.

|                              | (1)      |      | (2)    |      | (3)    |      | (4)    |      | (5)    |      | (6)          |      |
|------------------------------|----------|------|--------|------|--------|------|--------|------|--------|------|--------------|------|
|                              | Fin & In |      | Manuf  |      | Other  |      | Serv   |      | Transp |      | Whole/Retail |      |
|                              | mean     | p50  | mean   | p50  | mean   | p50  | mean   | p50  | mean   | p50  | mean         | p50  |
| Util. Exposure (USD million) | 49.8     | 16.6 | 16.4   | 5.28 | 45.4   | 17.1 | 19.4   | 4.66 | 32.6   | 15   | 12.4         | 3.72 |
| Maturity (years)             | 6.84     | 4.84 | 5.15   | 4.60 | 5.86   | 5.00 | 4.71   | 4.25 | 8.13   | 5.59 | 5.07         | 4.46 |
| Prob. of default (pct)       | 1.21     | 0.64 | 2.04   | 0.71 | 1.94   | 0.66 | 1.39   | 0.64 | 2.41   | 0.67 | 2.26         | 1.10 |
| Interest rate (pct)          | 2.47     | 1.92 | 3.78   | 2.98 | 3.48   | 2.98 | 2.69   | 2.01 | 3.51   | 3.12 | 4.09         | 3.25 |
| Loans newly past due (pct)   | 0.41     | 0    | 0.081  | 0    | 0.22   | 0    | 0.49   | 0    | 0.18   | 0    | 0.13         | 0    |
| Loans in for. currency (pct) | 0.73     | 1    | 0.56   | 1    | 0.58   | 1    | 0.58   | 1    | 0.63   | 1    | 0.49         | 0    |
| Observations                 | 25,488   |      | 50,696 |      | 22,884 |      | 19,107 |      | 5557   |      | 23,371       |      |

The table shows summary statistics of the baseline sample with 147,103 observations grouped by industry. Column 1: Finance & Insurance (naics 52) column 2: Manufacturing (naics 31–33), column 3: All other, column 4: Service Industries (naics 54–56, 61, 62, 71, 72, 81), column 5: Transportation and Warehousing (naics 42, 44, 45), column 6: Wholesale & Retail Trade (naics 48, 49). The table displays the means and the medians of the following variables: utilized exposure, maturity, bank-internal probability of default, interest rate, a dummy variable which is one when the borrower becomes past due on its loans in period  $t$ , and the share of a borrower's loans that are not denominated in the currency of the country where the borrower is located.

## Appendix B. Data appendix

### B.1. Y-14 data

There are several cleaning procedures which we apply to the raw Y-14 data to obtain the borrower-level dataset. Among other steps, we drop observations with missing customer id or loan id, loans with a D (default) rating that are not declared past due, and loans where the currency changed over time. A firm is identified as a combination of a unique customer id, bank id and country. We drop loans with zero utilized exposure.

### B.2. Other data sources

- USD Index: Trade-weighted “broad” USD index, calculated by Federal Reserve Board staff.
- Average share of foreign sales by country, industry and quarter: Thomson Reuters Worldscope.
- Exchange rates: Quarterly data from the International Financial Statistics (IFS) provided by the IMF.

- Exchange rate volatility: Quarterly exchange rate volatility calculated as the standard deviation of daily exchange rates within a quarter.
- Oil price: Downloaded from Haver.
- Inflation: Consumer Price Inflation from the International Financial Statistics (IFS) provided by the IMF.
- Exchange rate expectations: 24-month ahead forecast from the Survey of Professional Forecasters.

## Appendix C. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jinteco.2021.103558>.

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